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Varrato

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(54) **TIMING OF HAPTIC SIGNALS**
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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 413 days.

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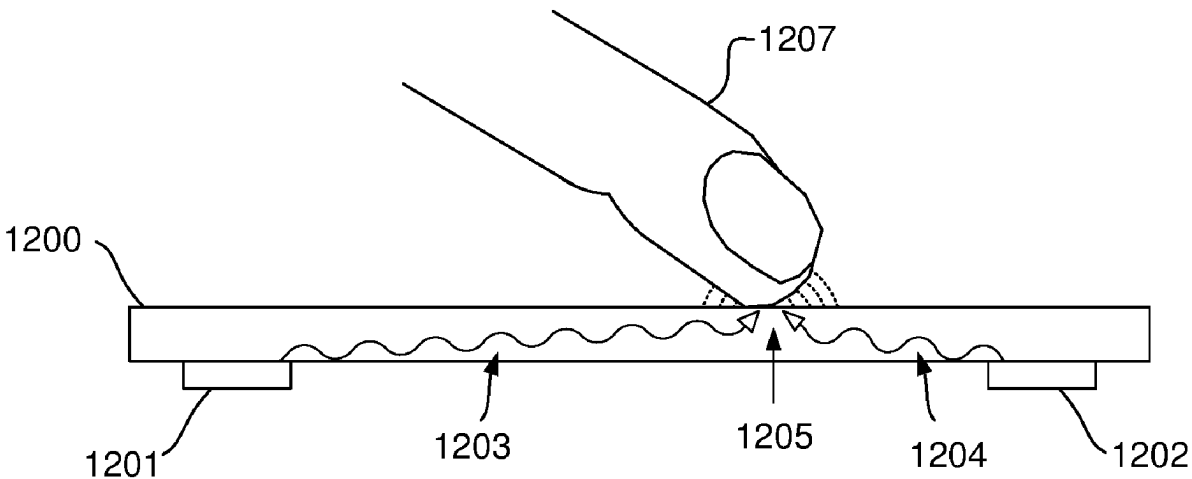
Primary Examiner — Duane N Taylor, Jr.

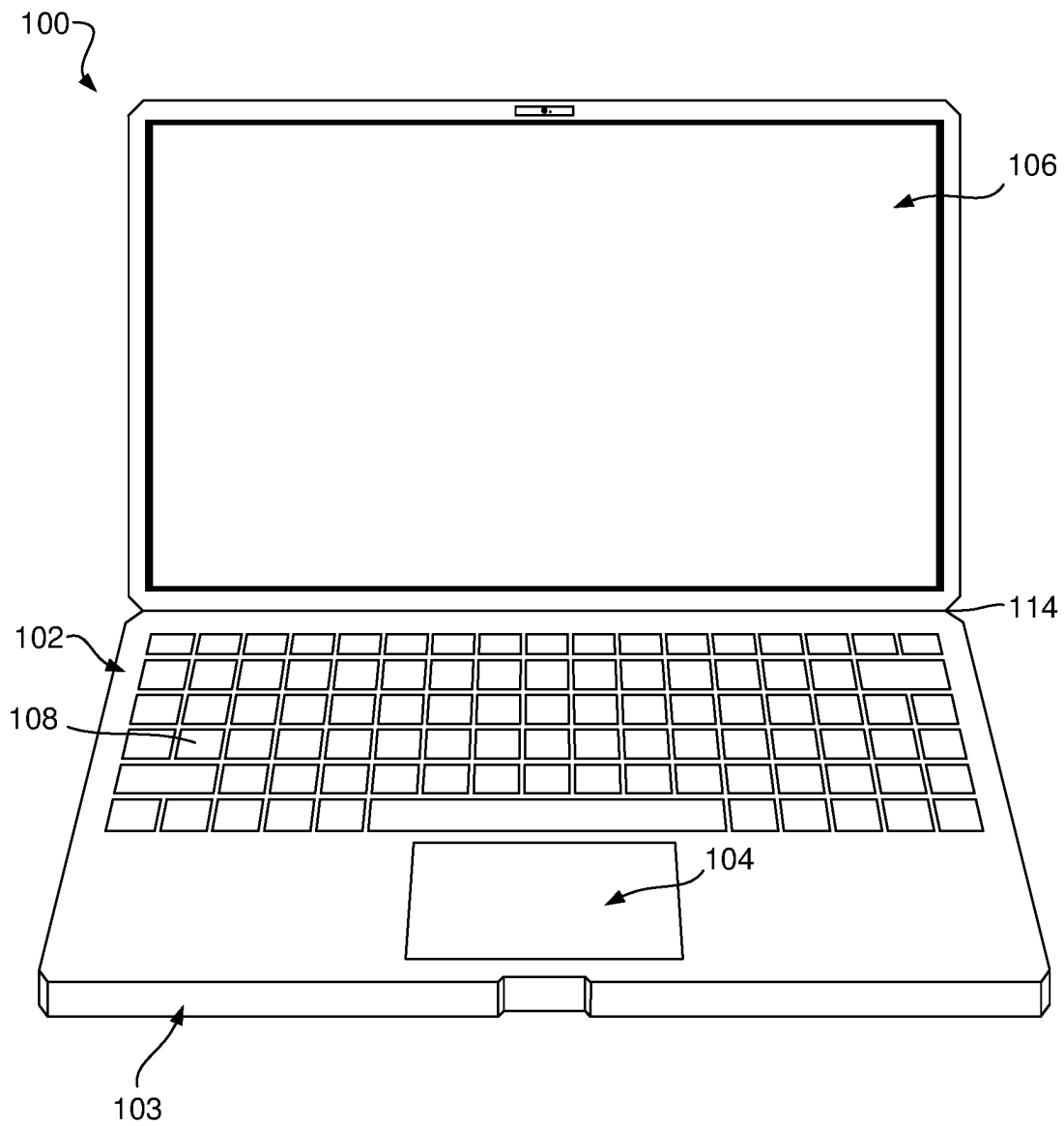
(57) **ABSTRACT**
Delivering a haptic feedback using a capacitance module may include obtaining a location of user input using a capacitive sensor in the capacitance module, determining when to activate a first haptic motor in the capacitance module based on the location of the user input, determining when the activate a second haptic motor in the capacitance module based on the location of the user input, generating a first haptic signal with the first haptic motor at the first determined time, and generating a second haptic signal with the second haptic motor at the second determined time, where the haptic feedback includes the first haptic signal and the second haptic signal.

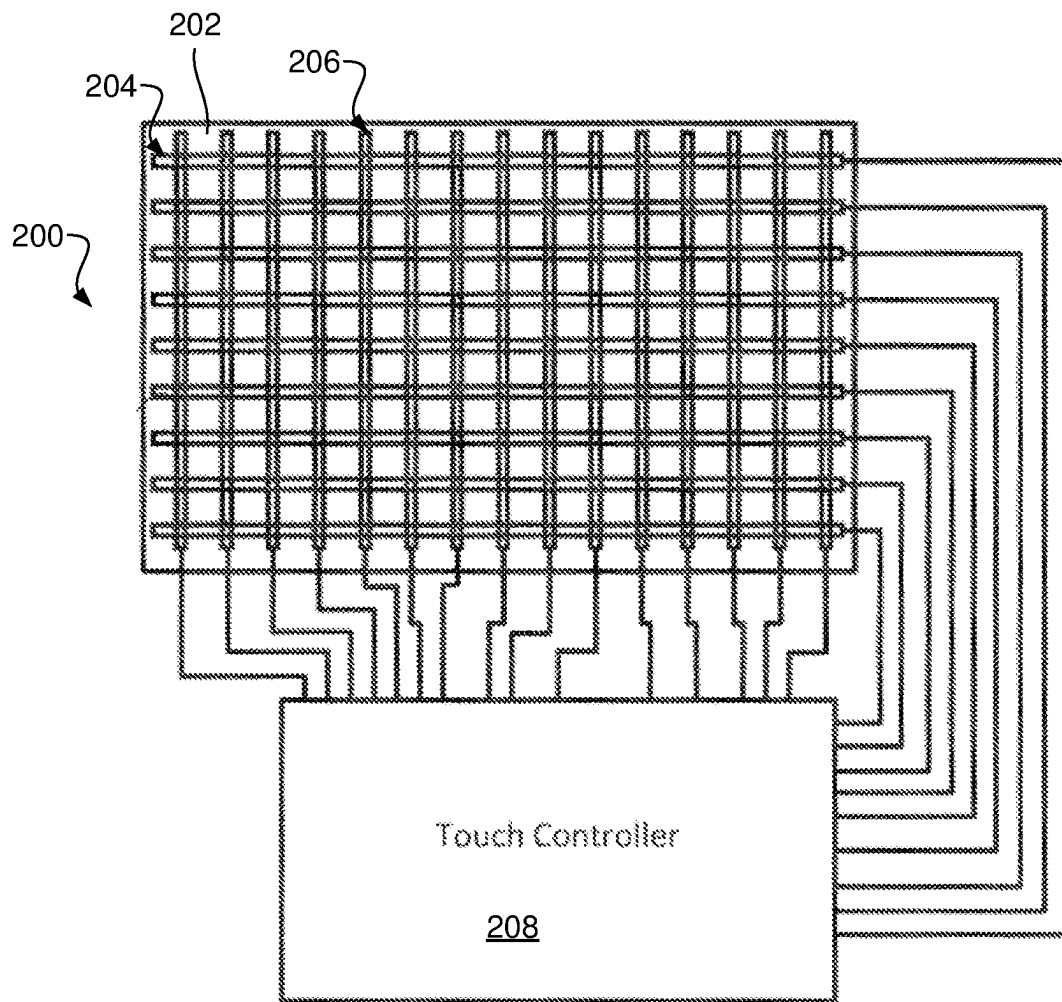
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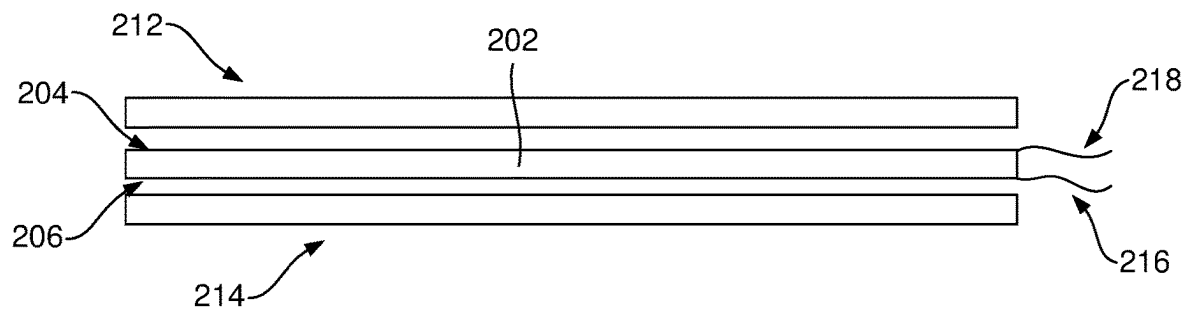
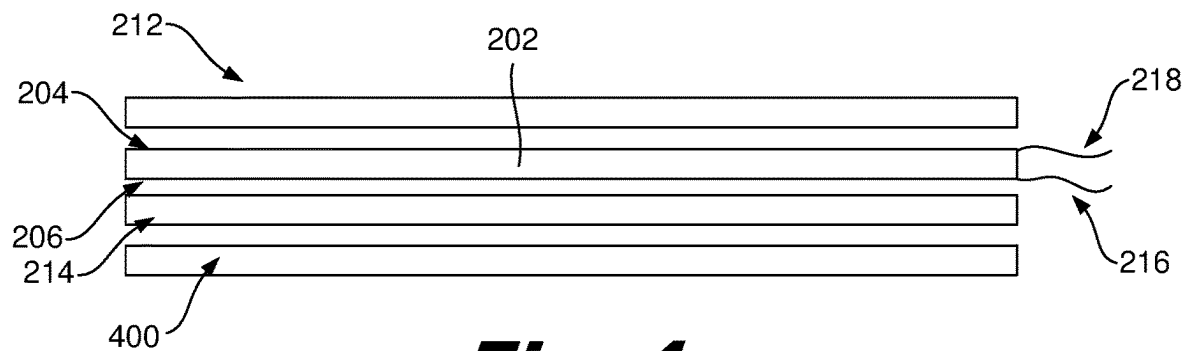
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G06F 3/0354 (2013.01)
G06F 3/044 (2006.01)
(52) **U.S. Cl.**
CPC **G06F 3/016** (2013.01); **G06F 3/03547** (2013.01); **G06F 3/0446** (2019.05)
(58) **Field of Classification Search**
None
See application file for complete search history.

18 Claims, 18 Drawing Sheets



***Fig. 1***

***Fig. 2***

**Fig. 3****Fig. 4**

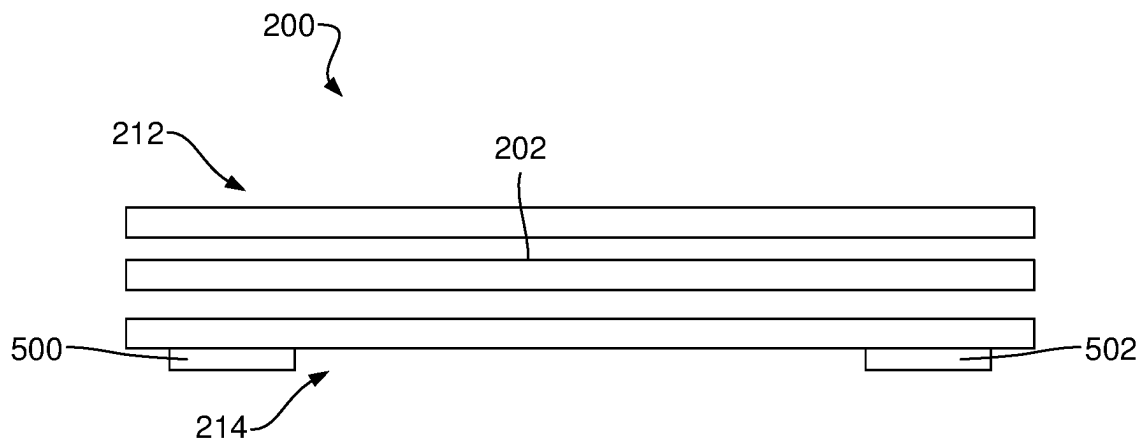


Fig. 5

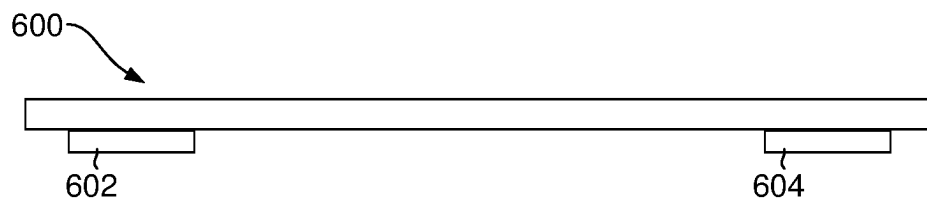


Fig. 6

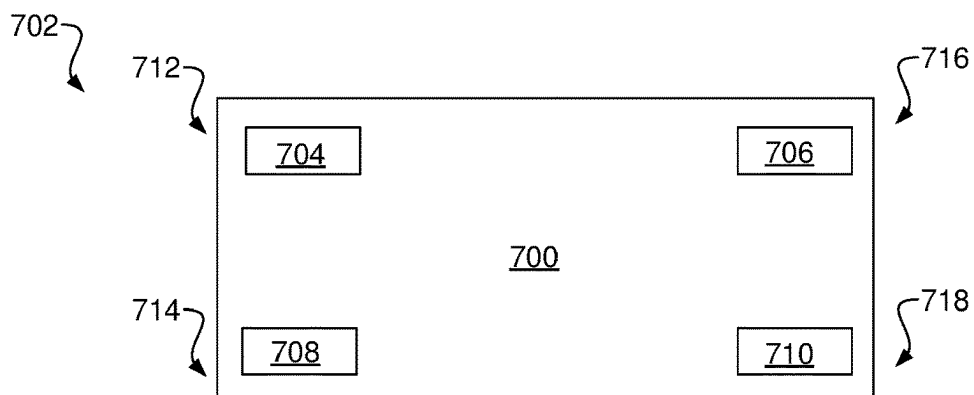


Fig. 7

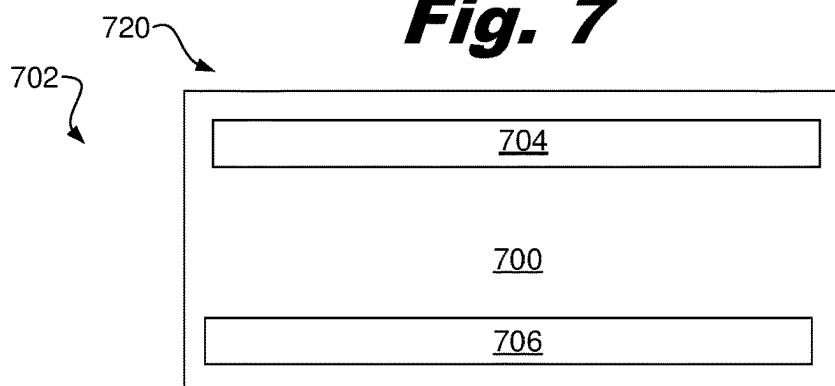


Fig. 8

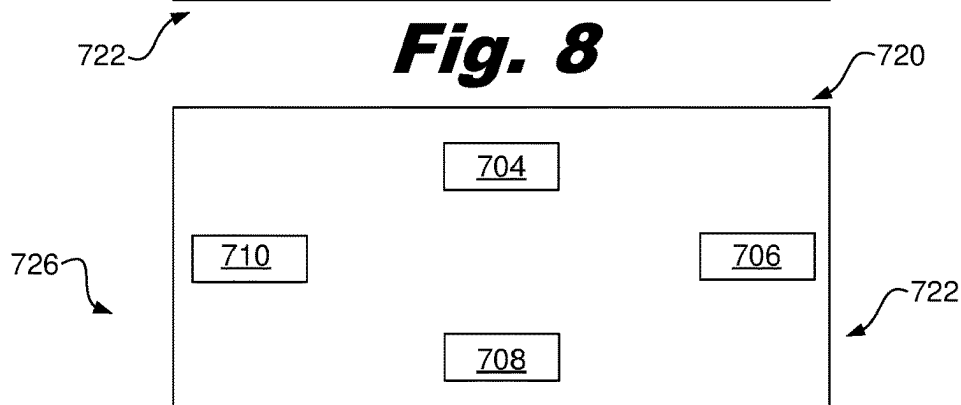


Fig. 9

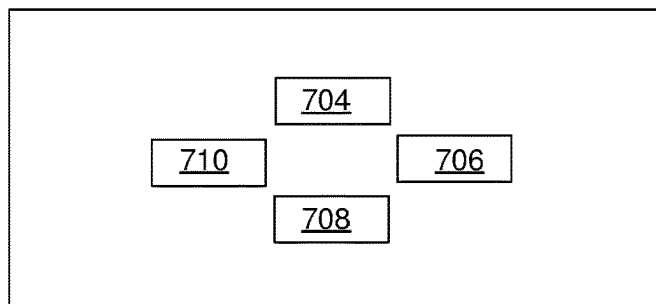


Fig. 10

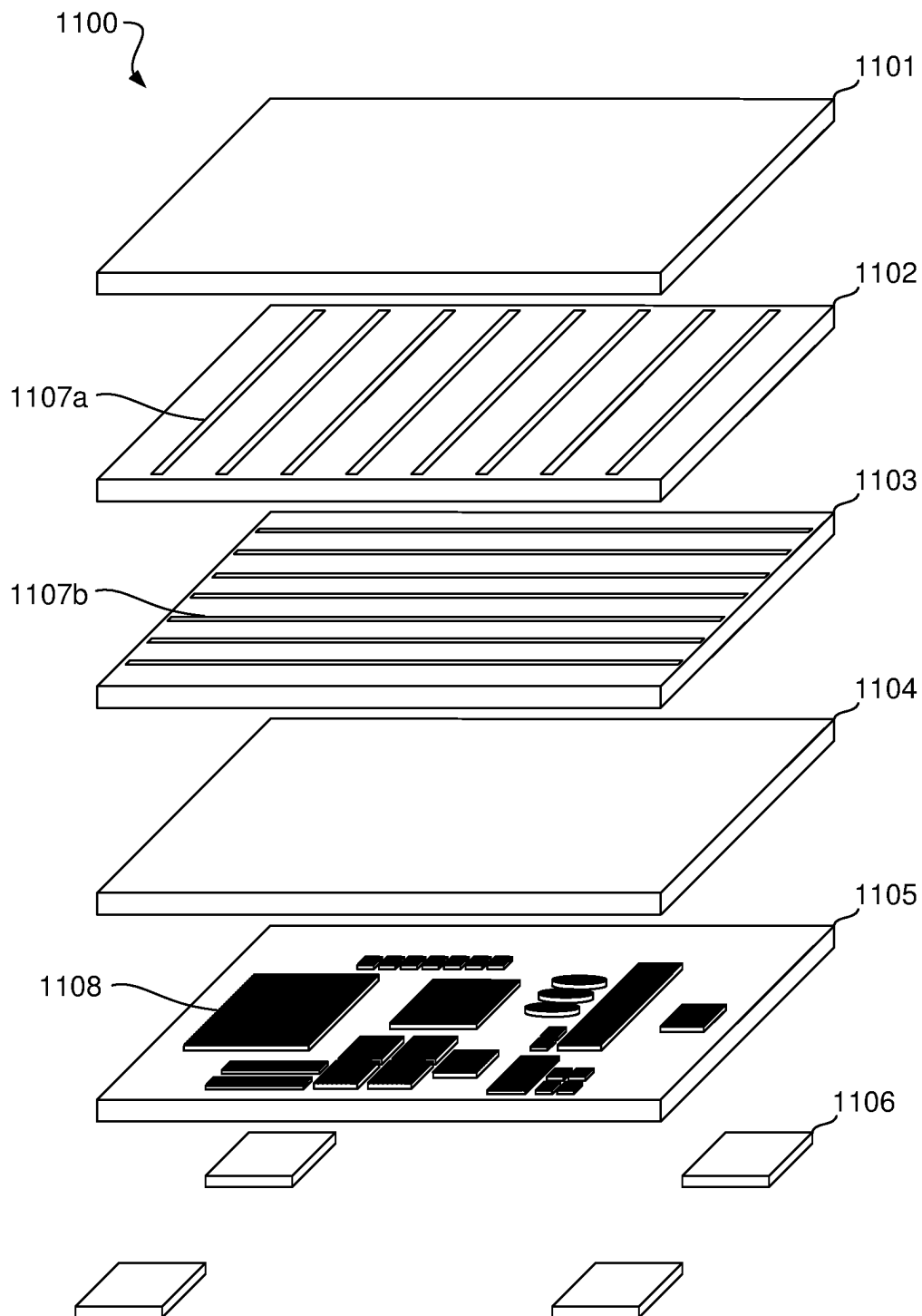
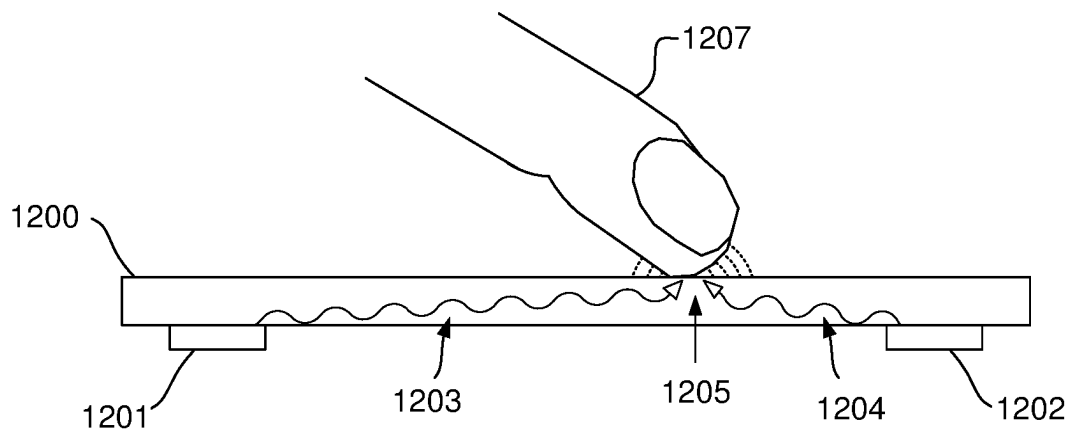
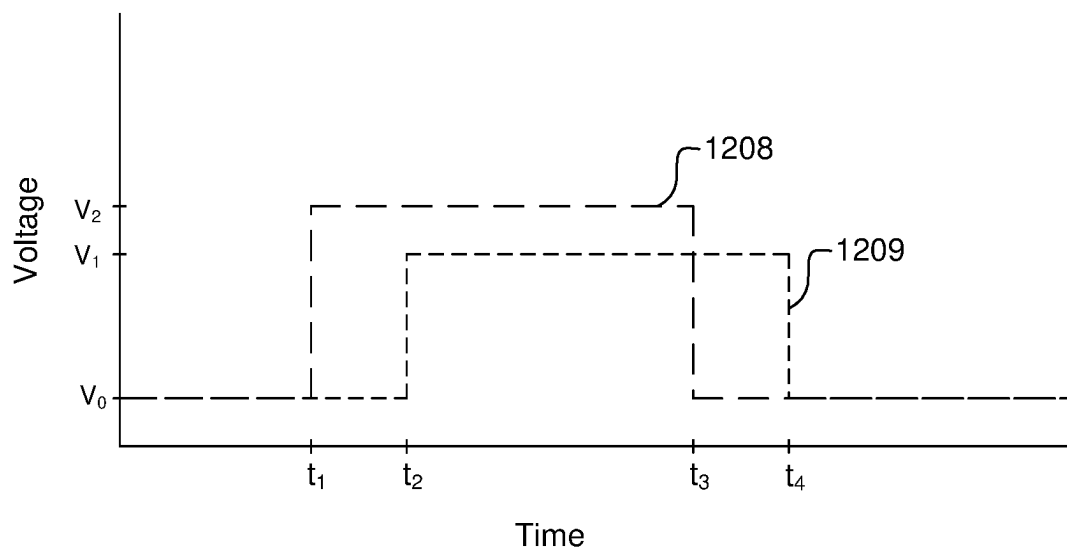
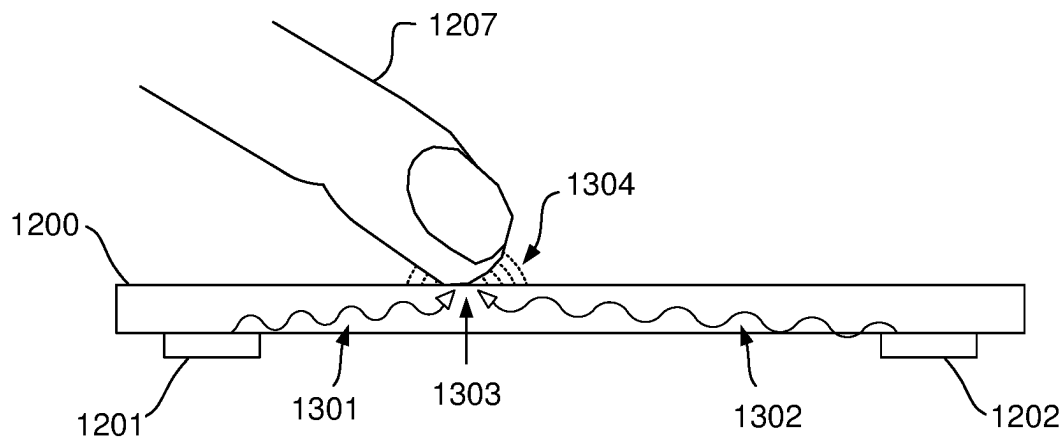
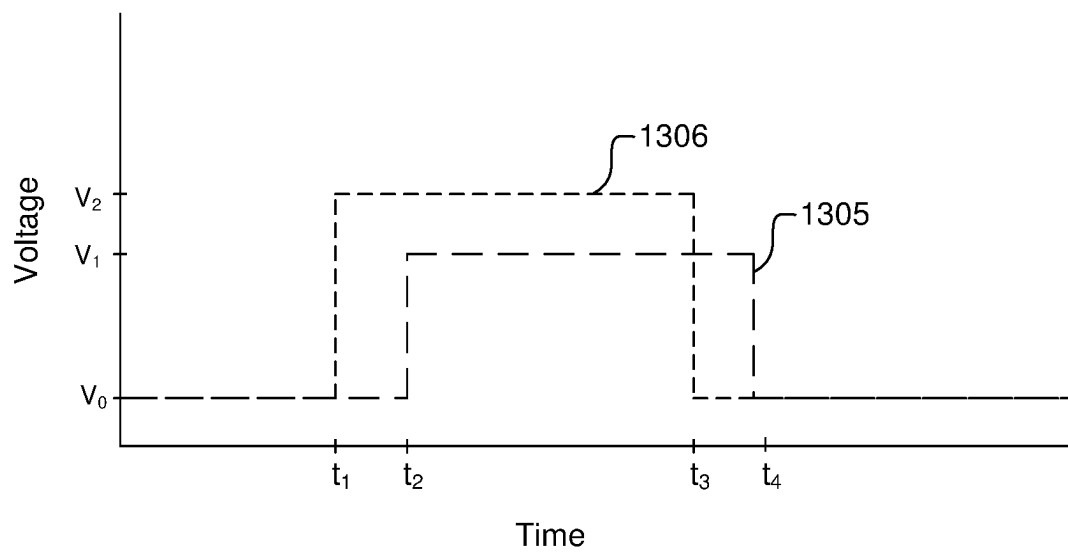
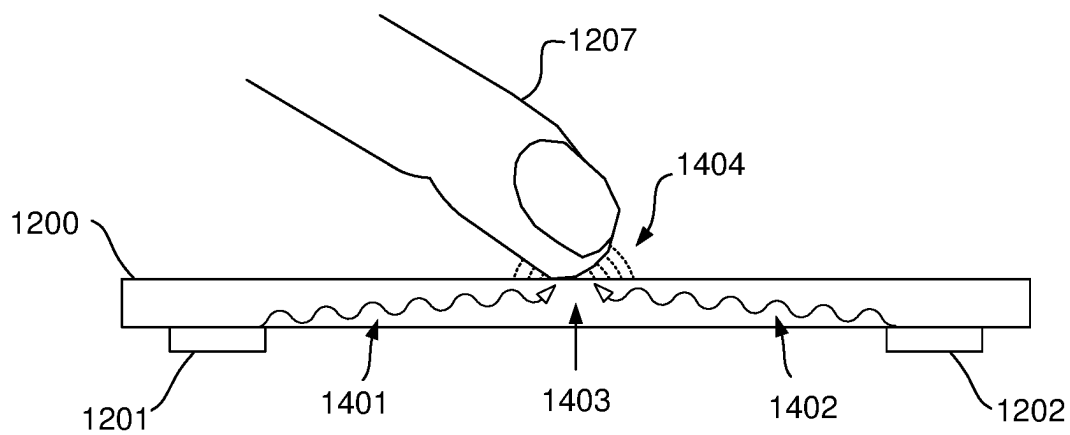
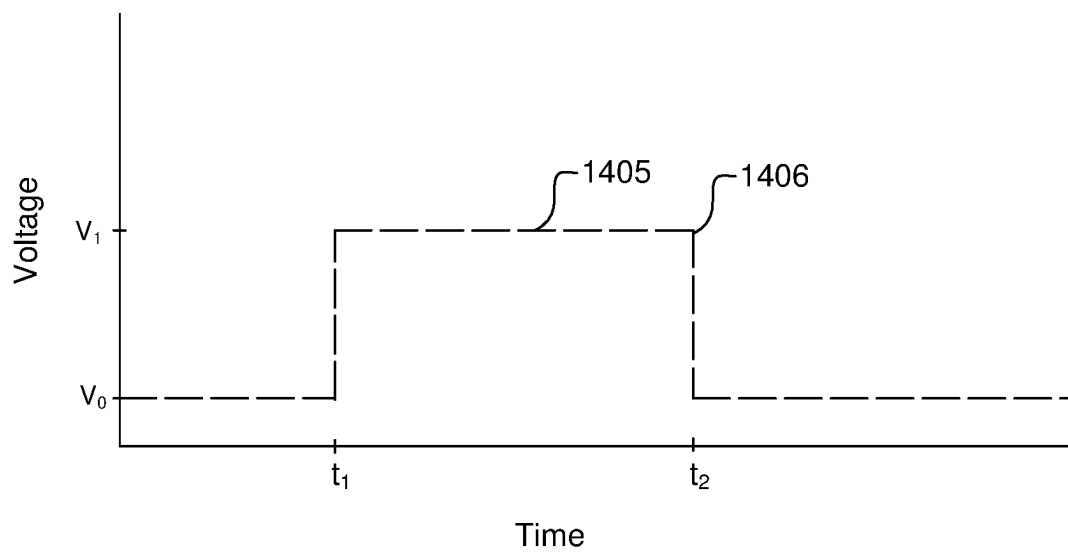
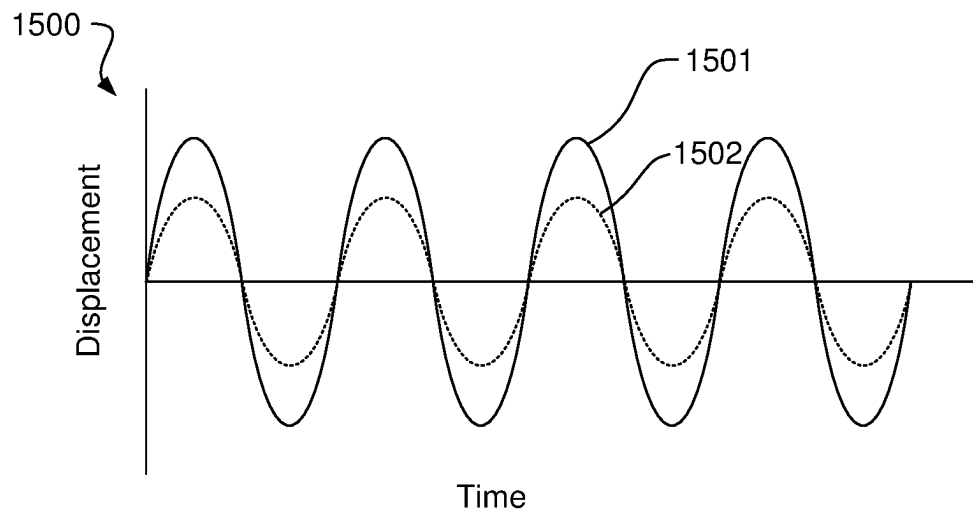
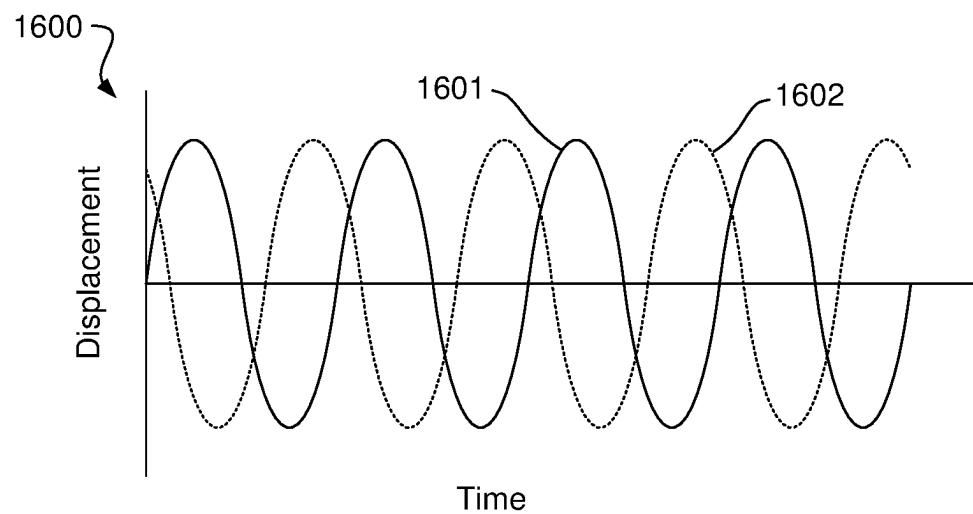


Fig. 11

**Fig. 12a****Fig. 12b**

**Fig. 13a****Fig. 13b**

***Fig. 14a******Fig. 14b***

***Fig. 15******Fig. 16***

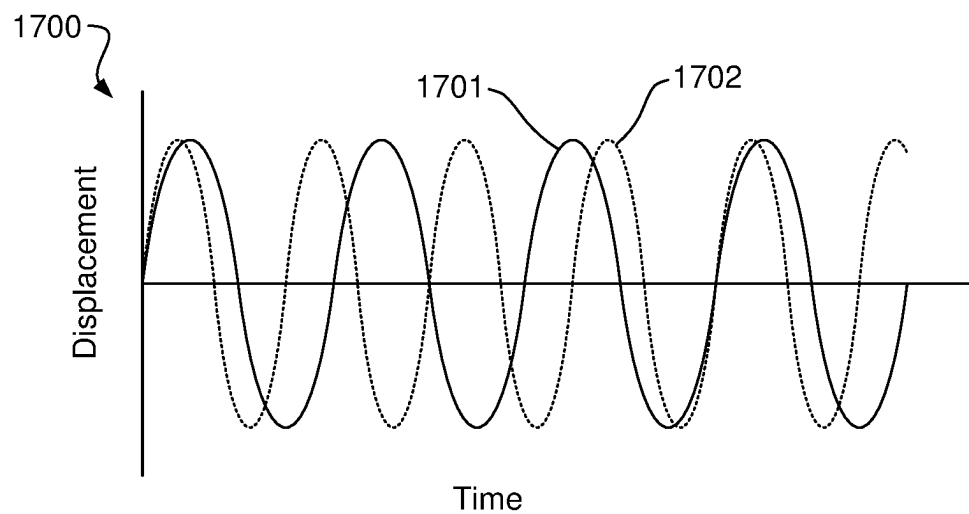


Fig. 17

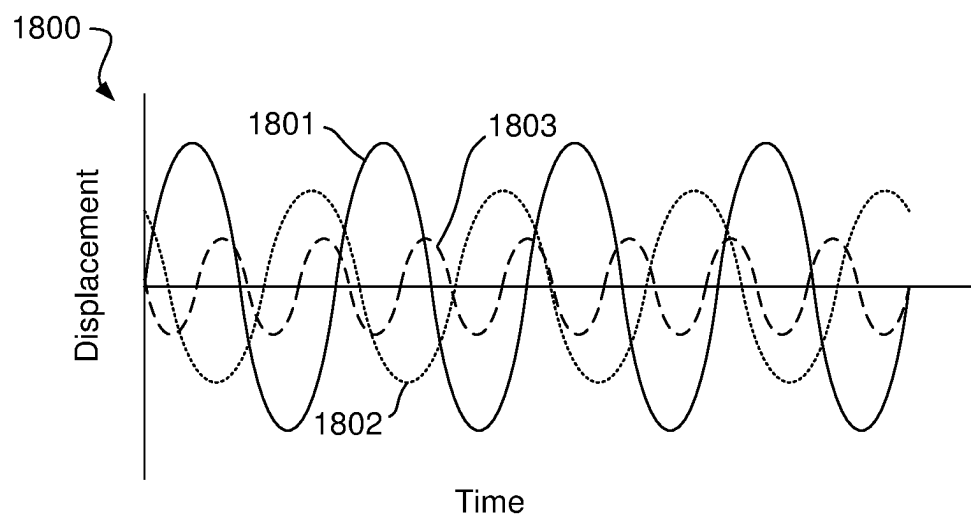


Fig. 18

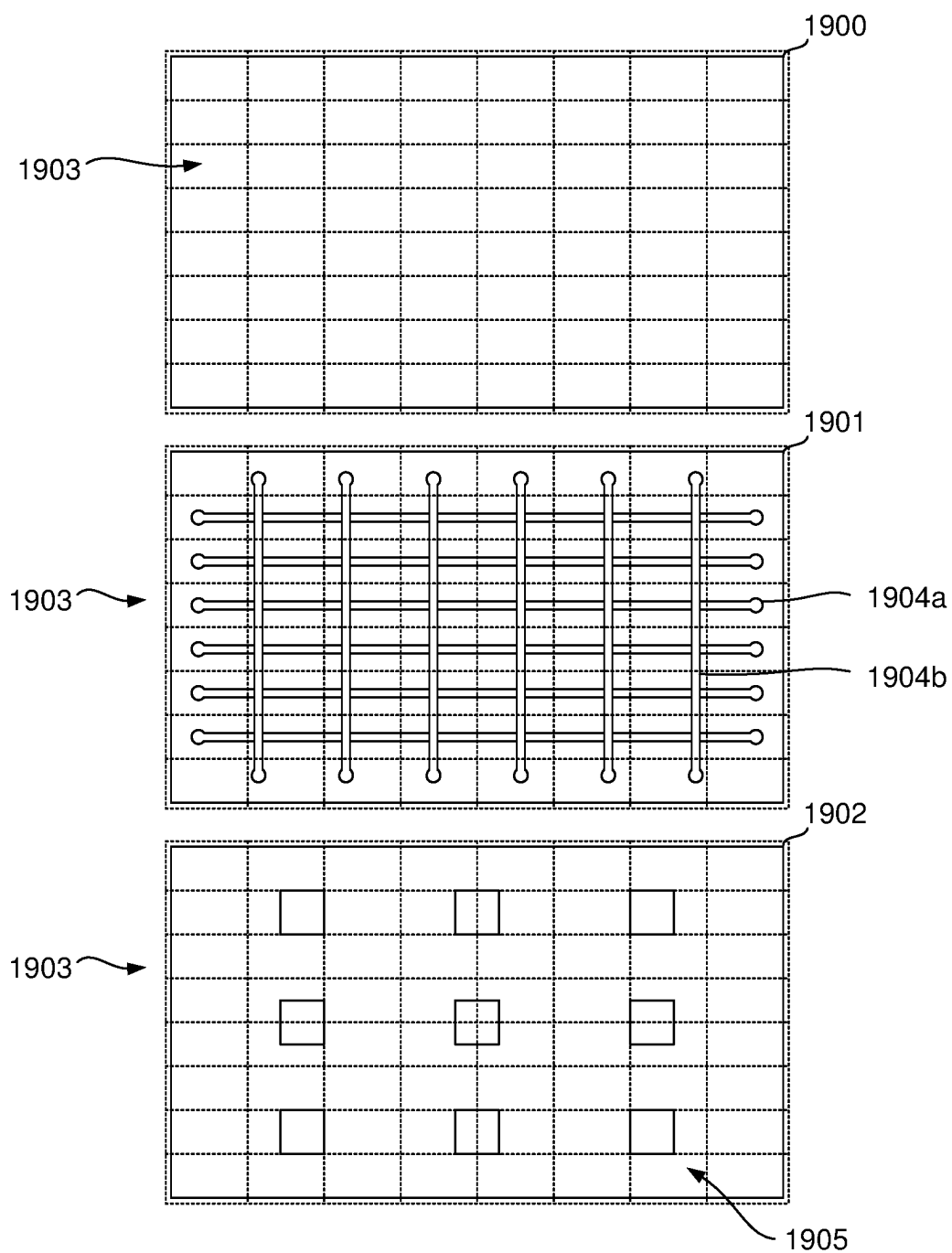


Fig. 19

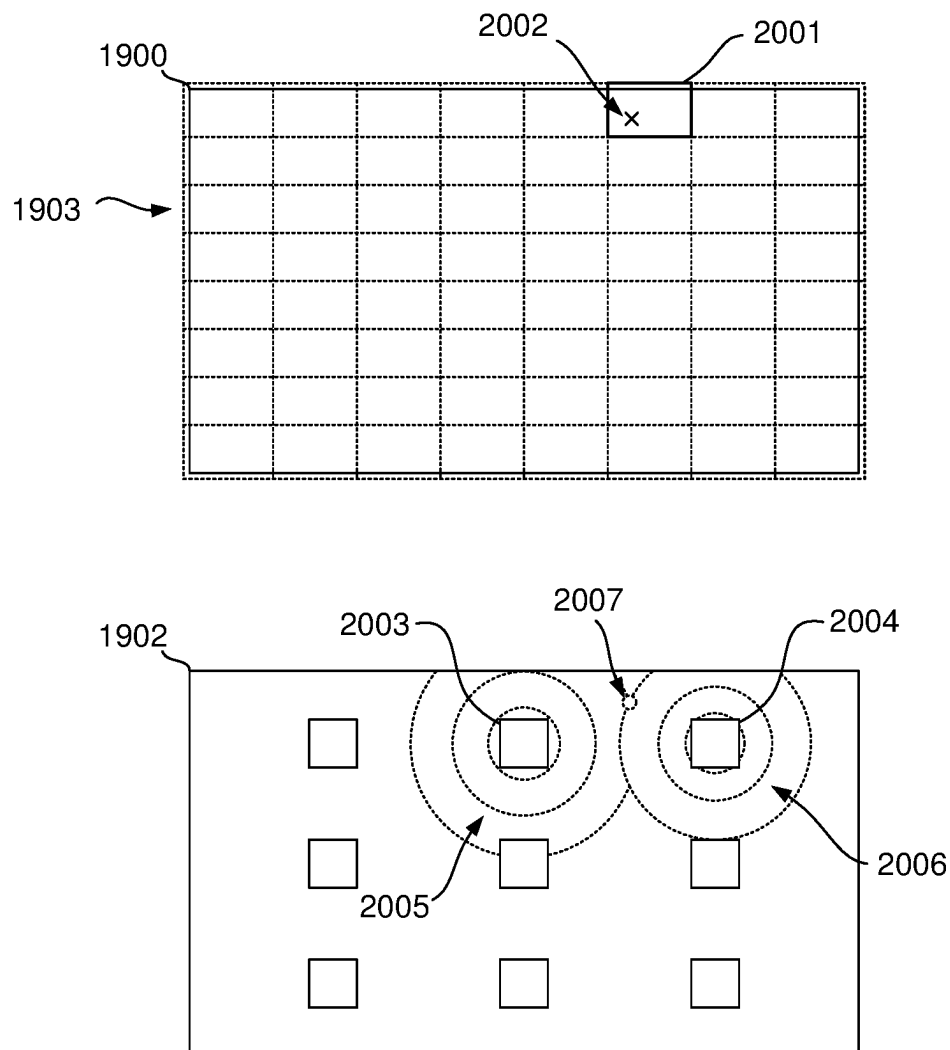


Fig. 20

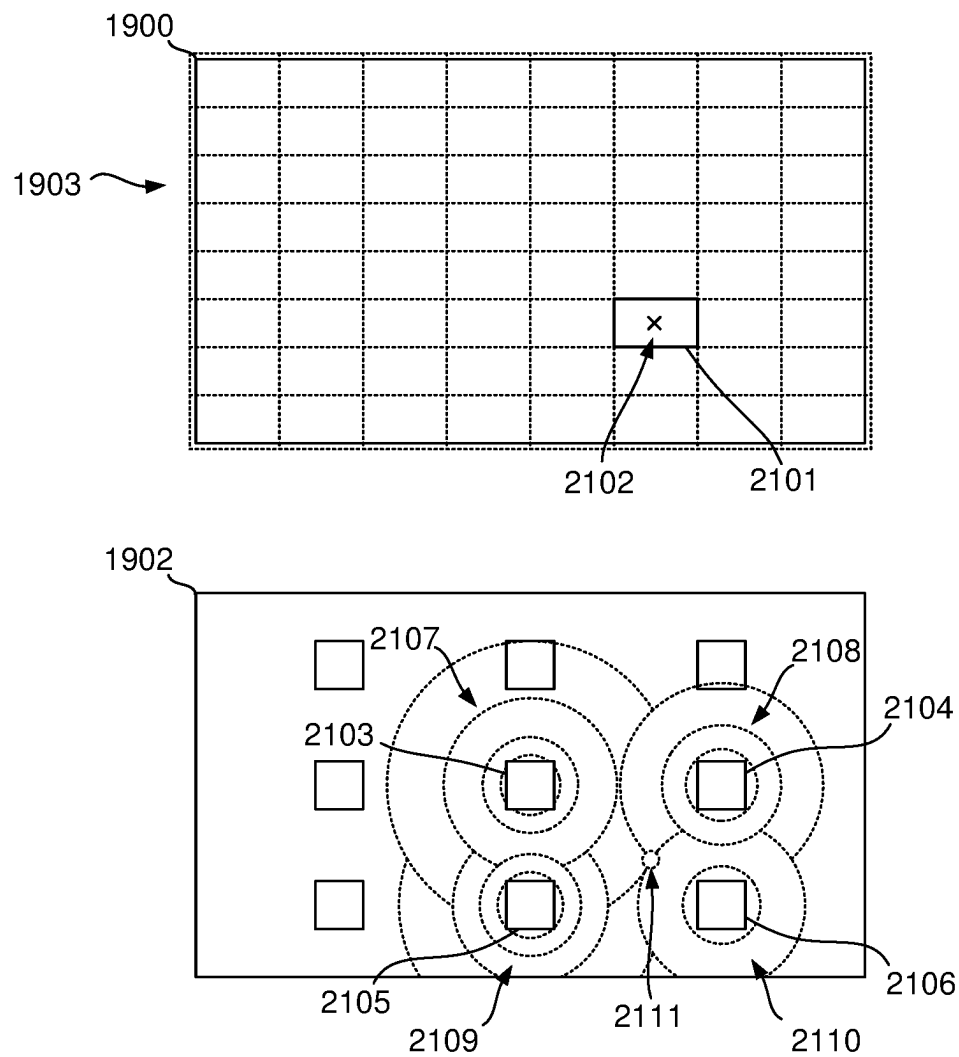


Fig. 21

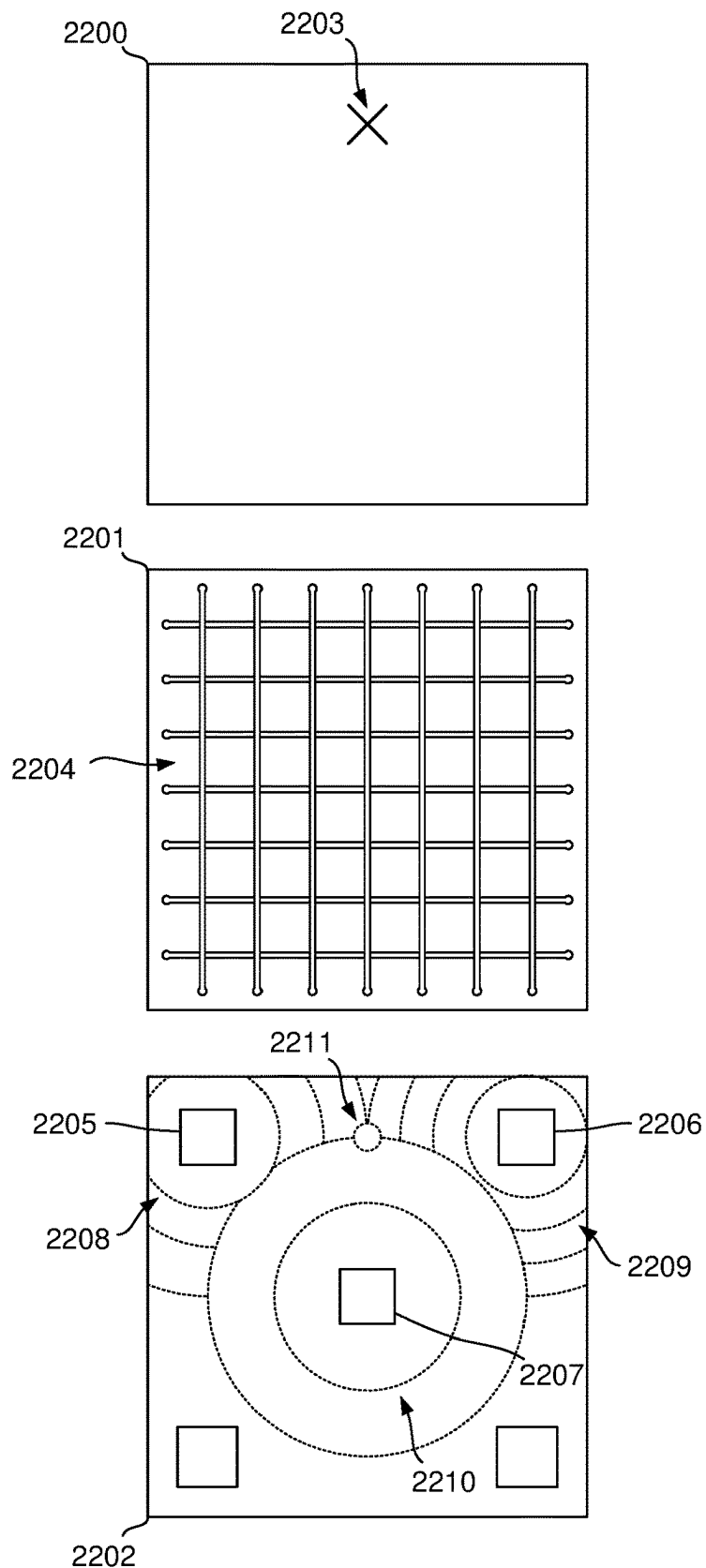


Fig. 22

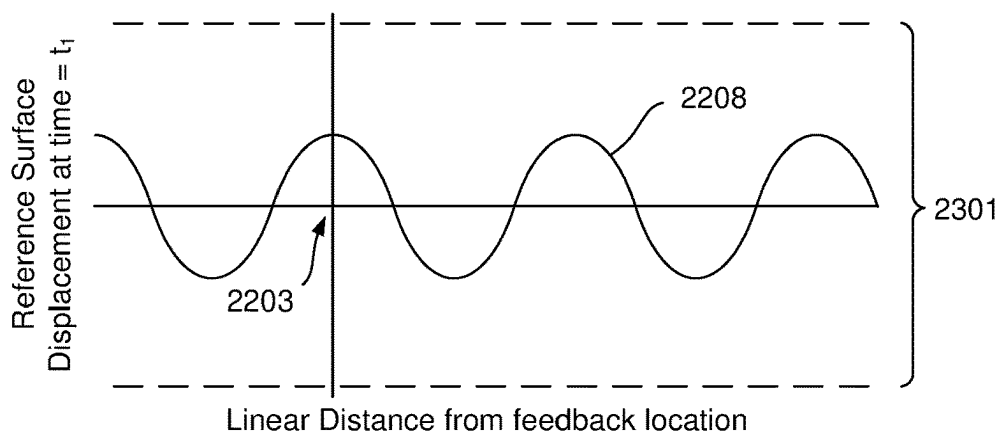


Fig. 23a

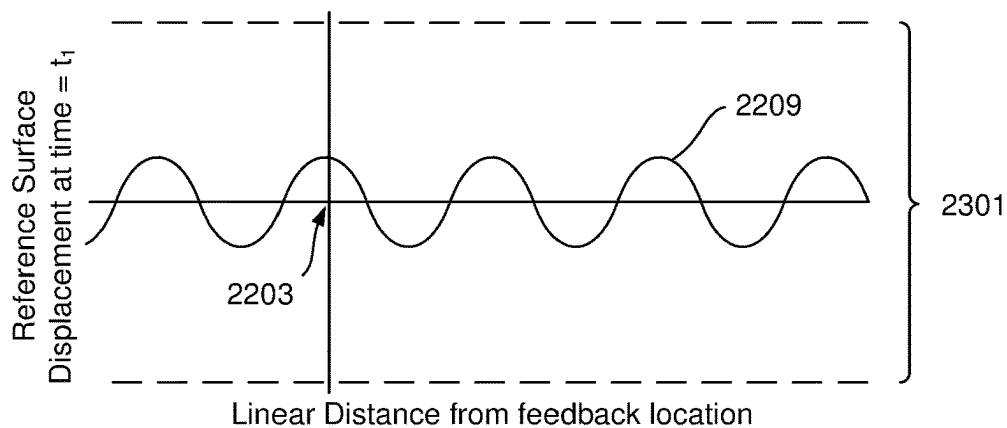


Fig. 23b

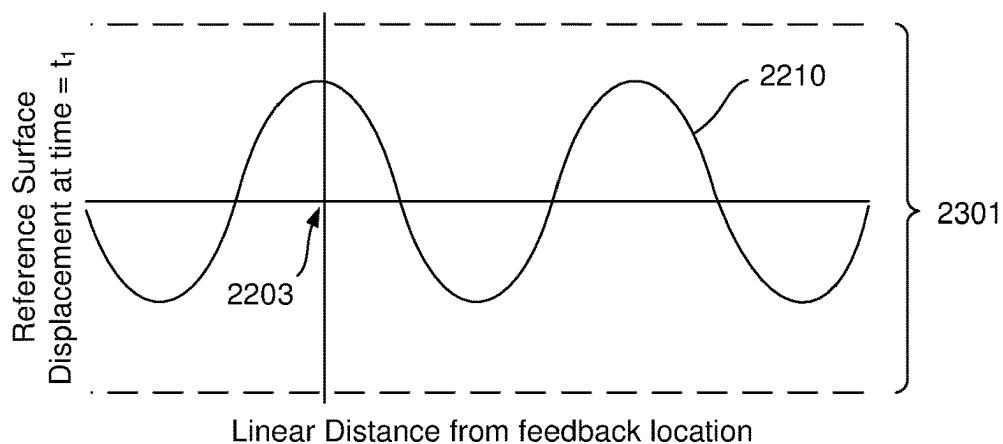


Fig. 23c

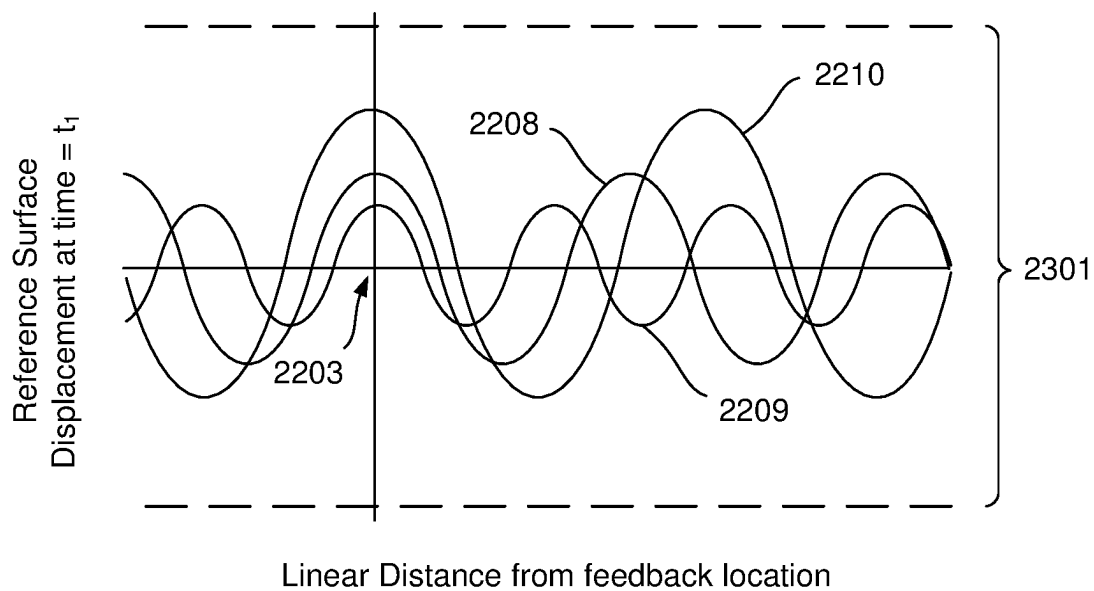


Fig. 23d

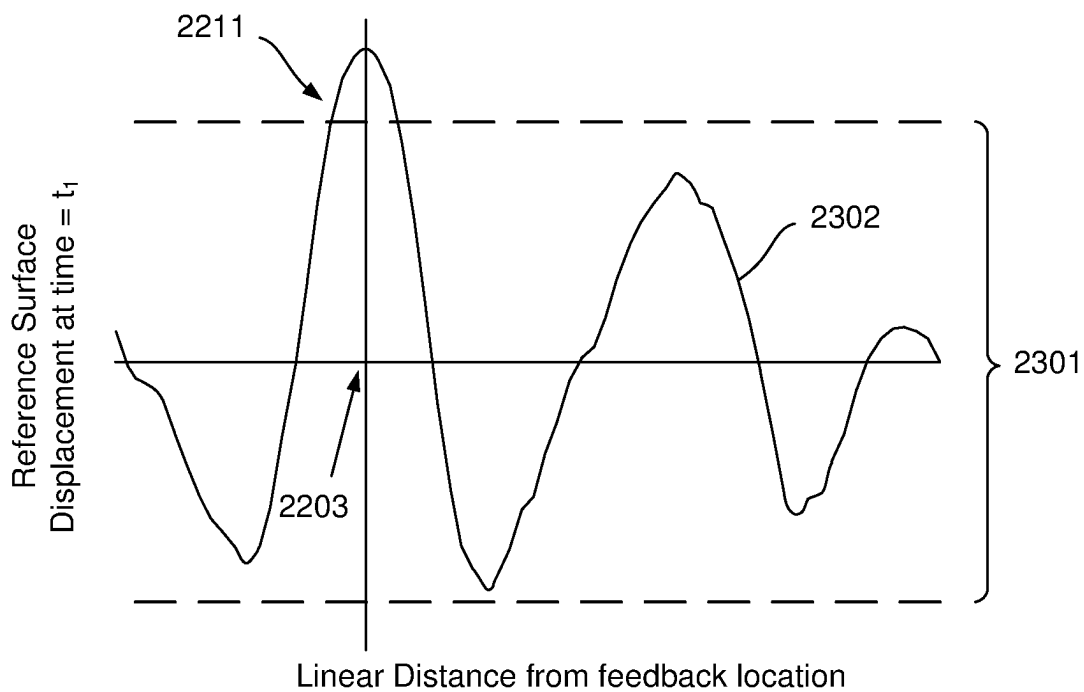
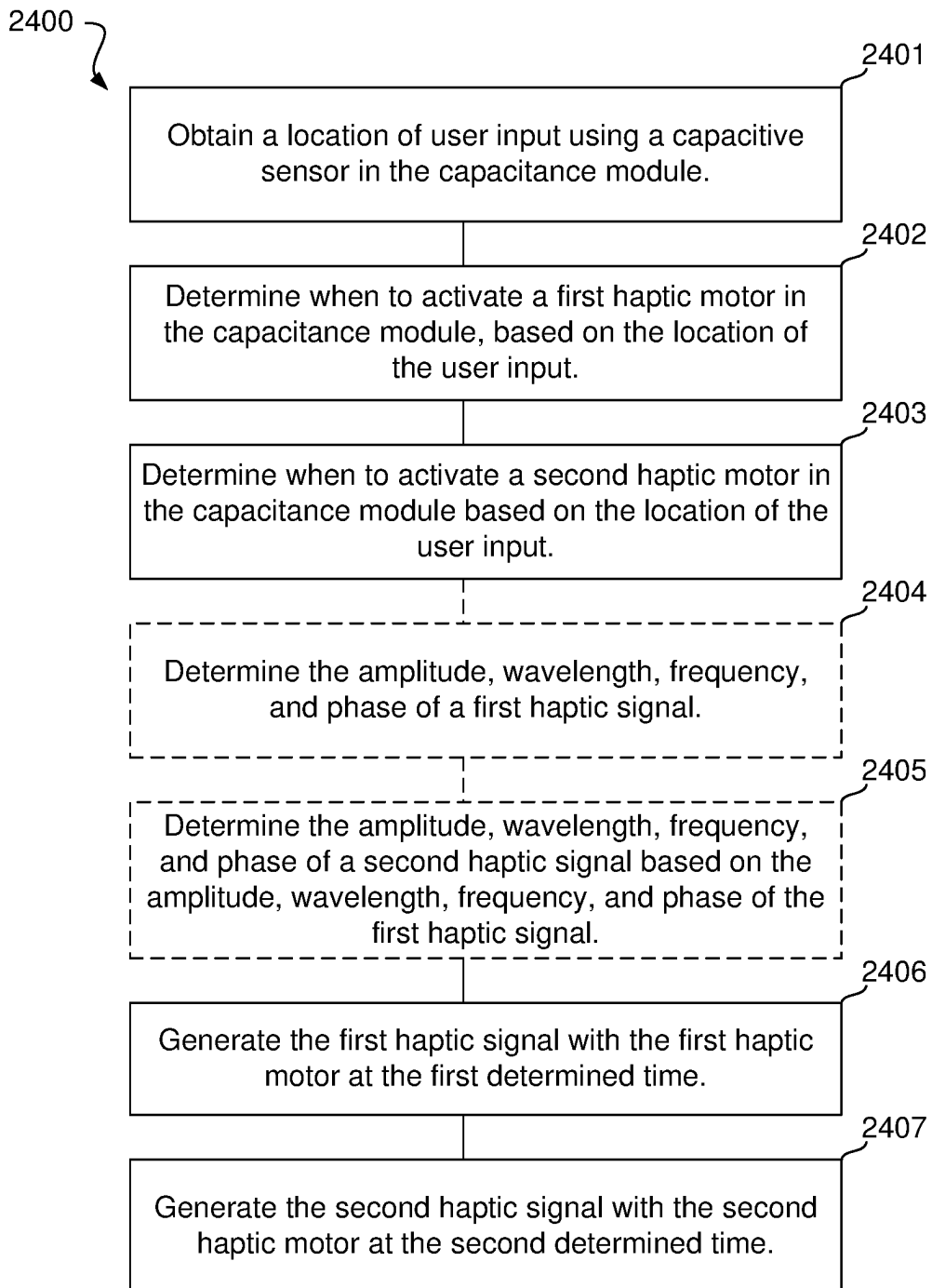


Fig. 23e

**Fig. 24**

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TIMING OF HAPTIC SIGNALS

FIELD OF THE DISCLOSURE

This disclosure relates generally to systems and methods for capacitance modules. In particular, this disclosure relates to systems and methods for generating haptic feedback using a plurality of haptic motors in a capacitance module.

BACKGROUND

Some devices include a haptic motor located underneath the touchable surface to deliver a haptic feedback that simulates a click. To deliver the haptic feedback, the haptic motor vibrates for a short duration of time. The haptic motor must vibrate with enough strength so that the vibration is felt by the user. Some devices include more than one haptic motor. By including more than one haptic motor, a more localized haptic feedback can be delivered. Multiple haptic motors consume more power than devices with a single haptic motor. In some applications, the problem of increasing power consumption may be proportional to the number of haptic motors used.

Some methods exist to maximize the efficiency of feedback devices that include haptic motors. Some methods include placing passive elements within a device to maximize vibrations at certain locations. When these passive elements receive a weak vibrational feedback from a haptic motor within the device, they create a stronger vibration that may be easier to be perceived by a user.

An example using a haptic motor to deliver feedback is disclosed in U.S. Pat. No. 8,633,916 issued to Jeffrey Traer Bernstein, et al. This example discloses a touch pad with force sensors and actuator feedback. Electronic devices may use touch pads that have touch sensor arrays, force sensors, and actuators for providing tactile feedback. A touch pad may be mounted in a computer housing. The touch pad may have a rectangular planar touch pad member that has a glass layer covered with ink and contains a capacitive touch sensor array. Force sensors may be used to measure how much force is applied to the surface of the planar touch pad member by a user. Processed force sensor signals may indicate the presence of button activity such as press and release events. In response to detected button activity or other activity in the device, actuator drive signals may be generated for controlling the actuator. The user may supply settings to adjust signal processing and tactile feedback parameters.

An example of using haptic feedback is disclosed in U.S. Pat. No. 11,209,931 issued to Lapoe E. Lynn, et al. This example discloses a haptic feedback system. The system includes a plurality of remote transmitters that are remote from a location of interest on a surface of the system. The system includes a signal generator that generates a signal for each of the remote transmitters. The remote transmitters propagate the signals through a medium of the surface and the signals interfere at the location of interest such that a localized disturbance is generated at the location of interest.

Yet another example of using haptic motors is disclosed in U.S. Pat. No. 11,030,865 issued to Massimiliano Di Luca, et al. This example discloses a haptic device including an active element and a plurality of passive elements coupled to the active element. Each passive element has a respective response to a wave transmitted by the active element, and is configured to cause a respective haptic effect at a respective location away from the active element. In some embodiments, the respective response includes resonating when

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energy having a resonant frequency is received. In some embodiments, the wave transmitted by the active element is in the form of a vibration or acoustic energy.

Each of these references are herein incorporated by reference for all that they disclose.

SUMMARY

In one embodiment, a capacitance module may include a capacitive sensor, a first haptic motor, a second haptic motor; wherein the first haptic motor and the second haptic motor are in mechanical communication with the capacitive sensor, and controlling resources connected to the capacitive sensor and the plurality of haptic motors with memory; wherein the memory contains programmed instructions that cause the controlling resources, when activated, to: determine when to activate the first haptic motor in response to a user input, independently determine when to activate the second haptic motor in response to the user input, generate a first haptic signal with the first haptic motor at a first determined time, and generate a second haptic signal with the second haptic motor at a second determined time.

The programmed instructions may cause the controlling resources, when activated, to: determine the amplitude, wavelength, frequency, and phase of the first haptic signal, and determine the amplitude, wavelength, frequency, and phase of the second haptic signal based, in part, on the amplitude, wavelength, frequency, and phase of the first haptic signal.

The first haptic signal and the second haptic signal may generate a haptic feedback that is just tangible at and/or near the location of user input.

The first haptic signal and second haptic signal may be timed to arrive at the location of user input at the same time.

The first haptic signal and second haptic signal may constructively interfere at the location of user input.

Determining when to activate the first haptic motor may include determining a distance between the user input and the first haptic motor and determining when to activate the second haptic motor may include determining a distance between the user input and the second haptic motor.

The capacitive sensor may include at least one pre-defined region, wherein the first determined time and second determined time are pre-programmed based on the pre-defined region corresponding to the location of user input.

The first determined time and the second determined time may have the same value.

The capacitance module may be a touchpad.

The first haptic signal and the second haptic signal may be tuned to minimize vibrations perceptible outside of the capacitance module.

In another embodiment, a method for delivering a haptic feedback with a capacitance module may include obtaining a location of user input using a capacitive sensor in the capacitance module, determining when to activate a first haptic motor in the capacitance module based on the location of the user input, independently determining when to activate a second haptic motor in the capacitance module based on the location of the user input, generating a first haptic signal with the first haptic motor at the first determined time, and generating a second haptic signal with the second haptic motor at the second determined time, wherein the haptic feedback comprises the first haptic signal and second haptic signal.

The method may include determining the amplitude, wavelength frequency, and phase of the first haptic signal, and determining the amplitude, wavelength, frequency, and

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phase of the second haptic signal based on the amplitude, wavelength, frequency, and phase of the first haptic signal.

The first haptic signal and second haptic signal may generate a haptic feedback the is just perceptible at and/or near the location of user input.

The first haptic signal and second haptic signal may be timed to arrive at the location of user input at the same time.

The first haptic signal and second haptic signal may constructively interfere at the location of user input.

Determining when to activate the first haptic motor may include determining a distance between the user input and the first haptic motor, and determining when to activate the second haptic motor may include determining a distance between the user input and the second haptic motor.

The capacitive sensor may include at least one pre-defined region, and the first determined time and second determined time may be pre-programmed based, in part, on the pre-defined region corresponding to the location of user input.

The capacitance module may be a touchpad.

In another embodiment, a computer-program product for generating a haptic feedback using a capacitance module, may include a non-transitory, computer readable medium storing instructions executable by a processor to: obtain a location of user input using a capacitive sensor in the capacitance module, determine when to activate a first haptic motor in the capacitance module based on the location of the user input, independently determine when to activate a second haptic motor in the capacitance module based on the location of the user input, generate a first haptic signal with the first haptic motor at the first determined time, and generate a second haptic signal with the second haptic motor at the second determined time, wherein the haptic feedback comprises the first haptic signal and second haptic signal.

The medium may store further instructions executable by a processor to: determine the amplitude, wavelength, frequency, and phase of the first haptic signal, and determine the amplitude, wavelength, frequency, and phase of the second haptic signal based on the amplitude, wavelength, frequency, and phase of the first haptic signal.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts an example of an electronic device in accordance with the disclosure.

FIG. 2 depicts an example of a substrate with a first set of electrodes and a second set of electrodes in accordance with the disclosure.

FIG. 3 depicts an example of a touch pad in accordance with the disclosure.

FIG. 4 depicts an example of a touch screen in accordance with the disclosure.

FIG. 5 depicts an example of an input surface in accordance with the disclosure.

FIG. 6 depicts an example of an input surface in accordance with the disclosure.

FIG. 7 depicts an example of an arrangement of pressure sensors in accordance with the disclosure.

FIG. 8 depicts an example of an arrangement of pressure sensors in accordance with the disclosure.

FIG. 9 depicts an example of an arrangement of pressure sensors in accordance with the disclosure.

FIG. 10 depicts an example of an arrangement of pressure sensors in accordance with the disclosure.

FIG. 11 depicts an example of a capacitance module in accordance with the disclosure.

FIG. 12a depicts an example of a haptic feedback in accordance with the disclosure.

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FIG. 12b depicts an example of a voltage graph in accordance with the disclosure.

FIG. 13a depicts an example of a haptic feedback in accordance with the disclosure.

FIG. 13b depicts an example of a voltage graph in accordance with the disclosure.

FIG. 14a depicts an example of a haptic feedback in accordance with the disclosure.

FIG. 14b depicts an example of a voltage graph in accordance with the disclosure.

FIG. 15 depicts an example of a displacement graph in accordance with the disclosure.

FIG. 16 depicts an example of a displacement graph in accordance with the disclosure.

FIG. 17 depicts an example of a displacement graph in accordance with the disclosure.

FIG. 18 depicts an example of a displacement graph in accordance with the disclosure.

FIG. 19 depicts an example of pre-programmed input regions in accordance with the disclosure.

FIG. 20 depicts an example of a haptic feedback in accordance with the disclosure.

FIG. 21 depicts an example of a haptic feedback in accordance with the disclosure.

FIG. 22 depicts an example of a user input on a capacitance module in accordance with the disclosure.

FIG. 23a depicts a displacement versus distance graph in accordance with the disclosure.

FIG. 23b depicts a displacement versus distance graph in accordance with the disclosure.

FIG. 23c depicts a displacement versus distance graph in accordance with the disclosure.

FIG. 23d depicts a displacement versus distance graph in accordance with the disclosure.

FIG. 23e depicts a displacement versus distance graph in accordance with the disclosure.

FIG. 24 depicts a method for delivering a haptic feedback with a capacitance module in accordance with the disclosure.

While the disclosure is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. However, it should be understood that the disclosure is not intended to be limited to the particular forms disclosed. Rather, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

This description provides examples, and is not intended to limit the scope, applicability or configuration of the invention. Rather, the ensuing description will provide those skilled in the art with an enabling description for implementing embodiments of the invention. Various changes may be made in the function and arrangement of elements.

Thus, various embodiments may omit, substitute, or add various procedures or components as appropriate. For instance, it should be appreciated that the methods may be performed in an order different than that described, and that various steps may be added, omitted, or combined. Also, aspects and elements described with respect to certain embodiments may be combined in various other embodiments. It should also be appreciated that the following systems, methods, devices, and software may individually or

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collectively be components of a larger system, wherein other procedures may take precedence over or otherwise modify their application.

For purposes of this disclosure, the term “aligned” generally refers to being parallel, substantially parallel, or forming an angle of less than 35.0 degrees. For purposes of this disclosure, the term “transverse” generally refers to perpendicular, substantially perpendicular, or forming an angle between 55.0 and 125.0 degrees. For purposes of this disclosure, the term “length” generally refers to the longest dimension of an object. For purposes of this disclosure, the term “width” generally refers to the dimension of an object from side to side and may refer to measuring across an object perpendicular to the object’s length.

For purposes of this disclosure, the term “electrode” may generally refer to a portion of an electrical conductor intended to be used to make a measurement, and the terms “route” and “trace” generally refer to portions of an electrical conductor that are not intended to make a measurement. For purposes of this disclosure in reference to circuits, the term “line” generally refers to the combination of an electrode and a “route” or “trace” portions of the electrical conductor. For purposes of this disclosure, the term “Tx” generally refers to a transmit line, electrode, or portions thereof, and the term “Rx” generally refers to a sense line, electrode, or portions thereof.

For the purposes of this disclosure, the term “electronic device” may generally refer to devices that can be transported and include a battery and electronic components. Examples may include a laptop, a desktop, a mobile phone, an electronic tablet, a personal digital device, a watch, a gaming controller, a gaming wearable device, a wearable device, a measurement device, an automation device, a security device, a display, a vehicle, an infotainment system, an audio system, a control panel, another type of device, an athletic tracking device, a tracking device, a card reader, a purchasing station, a kiosk, or combinations thereof.

It should be understood that use of the terms “capacitance module,” “touch pad” and “touch sensor” throughout this document may be used interchangeably with “capacitive touch sensor,” “capacitive sensor,” “capacitance sensor,” “capacitive touch and proximity sensor,” “proximity sensor,” “touch and proximity sensor,” “touch panel,” “trackpad,” “touch pad,” and “touch screen.”

It should also be understood that, as used herein, the terms “vertical,” “horizontal,” “lateral,” “upper,” “lower,” “left,” “right,” “inner,” “outer,” etc., can refer to relative directions or positions of features in the disclosed devices and/or assemblies shown in the Figures. For example, “upper” or “uppermost” can refer to a feature positioned closer to the top of a page than another feature. These terms, however, should be construed broadly to include devices and/or assemblies having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

In some cases, the capacitance module is located within a housing. The capacitance module may be underneath the housing and capable of detecting objects outside of the housing. In examples, where the capacitance module can detect changes in capacitance through a housing, the housing is a capacitance reference surface. For example, the capacitance module may be disclosed within a cavity formed by a keyboard housing of a computer, such as a laptop or other type of computing device, and the sensor may be disposed underneath a surface of the keyboard housing. In such an example, the keyboard housing adjacent to the

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capacitance module is the capacitance reference surface. In some examples, an opening may be formed in the housing, and an overlay may be positioned within the opening. In this example, the overlay is the capacitance reference surface. In such an example, the capacitance module may be positioned adjacent to a backside of the overlay, and the capacitance module may sense the presence of the object through the thickness of the overlay. For the purposes of this disclosure, the term “reference surface” may generally refer to a surface through which a pressure sensor, a capacitance sensor, or another type of sensor is positioned to sense a pressure, a presence, a position, a touch, a proximity, a capacitance, a magnetic property, an electric property, another type of property, or another characteristic, or combinations thereof that indicates an input. For example, the reference surface may be a housing, an overlay, or another type of surface through which the input is sensed. In some examples, the reference surface has no moving parts. In some examples, the reference surface may be made of any appropriate type of material, including, but not limited to, plastics, glass, a dielectric material, a metal, another type of material, or combinations thereof.

For the purposes of this disclosure, the term “display” may generally refer to a display or screen that is not depicted in the same area as the capacitive reference surface. In some cases, the display is incorporated into a laptop where a keyboard is located between the display and the capacitive reference surface. In some examples where the capacitive reference surface is incorporated into a laptop, the capacitive reference surface may be part of a touch pad. Pressure sensors may be integrated into the stack making up the capacitance module. However, in some cases, the pressure sensors may be located at another part of the laptop, such as under the keyboard housing, but outside of the area used to sense touch inputs, on the side of the laptop, above the keyboard, to the side of the keyboard, at another location on the laptop, or at another location. In examples where these principles are integrated into a laptop, the display may be pivotally connected to the keyboard housing. The display may be a digital screen, a touch screen, another type of screen, or combinations thereof. In some cases, the display is located on the same device as the capacitive reference surface, and in other examples, the display is located on another device that is different from the device on which the capacitive reference surface is located. For example, the display may be projected onto a different surface, such as a wall or projector screen. In some examples, the reference surface may be located on an input or gaming controller, and the display is located on a wearable device, such as a virtual reality or augmented reality screen. In some cases, the reference surface and the display are located on the same surface, but on separate locations on that surface. In other examples, the reference surface and the display may be integrated into the same device, but on different surfaces. In some cases, the reference surface and the display may be oriented at different angular orientations with respect to each other.

For the purposes of this disclosure, the term “haptic signal” may be generally understood as a vibration generated by a haptic motor. The vibration of a haptic motor may be modelled as a vertical displacement with respect to time. In some cases, the haptic signal may be understood as a propagating wave, which may be an accurate model over small intervals of time, typically intervals of microseconds. The vibration of a haptic motor may also be modelled as vertical displacement with respect to horizontal distance. In some cases, the haptic signal may be understood as a

standing wave, which may be more accurate over larger intervals of time, typically intervals of milliseconds.

For the purposes of this disclosure, the term “haptic feedback” may be generally understood as a vibration that is meant to be perceived by the user. A haptic feedback may be created by the combination of two or more haptic signals which are small enough in amplitude to be imperceptible on their own. In some cases, one or more of the combined signals may be perceptible on its own. In other examples, haptic feedback can be created by a single haptic signal that is perceptible on its own.

For purposes of this disclosure, a haptic motor may be generally understood as any appropriate device that generates a haptic signal. A non-exhaustive list of types of haptic motors includes, but is not limited to, piezoelectric devices, magnetostrictive devices, vibrators, eccentric rotating masses, linear resonant actuators, another type of actuator, another type of device, or combinations thereof.

In some cases, when haptic motors are included in a capacitance module, as a touchpad in a laptop, haptic feedback may be used to simulate the sensation of depressing the touchpad. While very little physical depression of the capacitance module may take place, the haptic feedback may provide feedback to the user that their input on the touchpad was registered as a click.

FIG. 1 depicts an example of an electronic device 100. In this example, the electronic device is a laptop. In the illustrated example, the electronic device 100 includes input components, such as a keyboard 102 and a capacitive module, such as a touch pad 104, that are incorporated into a housing 103. The electronic device 100 also includes a display 106. A program operated by the electronic device 100 may be depicted in the display 106 and controlled by a sequence of instructions that are provided by the user through the keyboard 102 and/or through the touch pad 104. An internal battery (not shown) may be used to power the operations of the electronic device 100.

The keyboard 102 includes an arrangement of keys 108 that can be individually selected when a user presses on a key with a sufficient force to cause the key 108 to be depressed towards a switch located underneath the keyboard 102. In response to selecting a key 108, a program may receive instructions on how to operate, such as a word processing program determining which types of words to process. A user may use the touch pad 104 to give different types of instructions to the programs operating on the computing device 100. For example, a cursor depicted in the display 106 may be controlled through the touch pad 104. A user may control the location of the cursor by sliding his or her hand along the surface of the touch pad 104. In some cases, the user may move the cursor to be located at or near an object in the computing device's display and give a command through the touch pad 104 to select that object. For example, the user may provide instructions to select the object by tapping the surface of the touch pad 104 one or more times.

The touch pad 104 is a capacitance module that includes a stack of layers disposed underneath the keyboard housing, underneath an overlay that is fitted into an opening of the keyboard housing, or underneath another capacitive reference surface. In some examples, the capacitance module is located in an area of the keyboard's surface where the user's palms may rest while typing. The capacitance module may include a substrate, such as a printed circuit board or another type of substrate. One of the layers of the capacitance module may include a sensor layer that includes a first set of electrodes oriented in a first direction and a second layer of

electrodes oriented in a second direction that is transverse the first direction. These electrodes may be spaced apart and/or electrically isolated from each other. The electrical isolation may be accomplished by deposited at least a portion of the electrodes on different sides of the same substrate or providing dedicated substrates for each set of electrodes. Capacitance may be measured at the overlapping intersections between the different sets of electrodes. However, as an object with a different dielectric value than the surrounding air (e.g., finger, stylus, etc.) approach the intersections between the electrodes, the capacitance between the electrodes may change. This change in capacitance and the associated location of the object in relation to the capacitance module may be calculated to determine where the user is touching or hovering the object within the detection range of the capacitance module. In some examples, the first set of electrodes and the second set of electrodes are equidistantly spaced with respect to each other. Thus, in these examples, the sensitivity of the capacitance module is the same in both directions. However, in other examples, the distance between the electrodes may be non-uniformly spaced to provide greater sensitivity for movements in certain directions.

In some cases, the display 106 is mechanically separate and movable with respect to the keyboard with a connection mechanism 114. In these examples, the display 106 and keyboard 102 may be connected and movable with respect to one another. The display 106 may be movable within a range of 0 degrees to 180 degrees or more with respect to the keyboard 102. In some examples, the display 106 may fold over onto the upper surface of the keyboard 102 when in a closed position, and the display 106 may be folded away from the keyboard 102 when the display 106 is in an operating position. In some examples, the display 106 may be orientable with respect to the keyboard 102 at an angle between 35 to 135 degrees when in use by the user. However, in these examples, the display 106 may be positionable at any angle desired by the user.

In some examples, the display 106 may be a non-touch sensitive display. However, in other examples at least a portion of the display 106 is touch sensitive. In these examples, the touch sensitive display may also include a capacitance module that is located behind an outside surface of the display 106. As a user's finger or other object approaches the touch sensitive screen, the capacitance module may detect a change in capacitance as an input from the user.

While the example of FIG. 1 depicts an example of the electronic device being a laptop, the capacitance sensor and touch surface may be incorporated into any appropriate device. A non-exhaustive list of devices includes, but is not limited to, a desktop, a display, a screen, a kiosk, a computing device, an electronic tablet, a smart phone, a location sensor, a card reading sensor, another type of electronic device, another type of device, or combinations thereof.

FIG. 2 depicts an example of a portion of a capacitance module 200. In this example, the capacitance module 200 may include a substrate 202, first set 204 of electrodes, and a second set 206 of electrodes. The first and second sets 204, 206 of electrodes may be oriented to be transverse to each other. Further, the first and second sets 204, 206 of electrodes may be electrically isolated from one another so that the electrodes do not short to each other. However, where electrodes from the first set 204 overlap with electrodes from the second set 206, capacitance can be measured. The capacitance module 200 may include one or more electrodes in the first set 204 or the second set 206. Such a substrate 202

and electrode sets may be incorporated into a touch screen, a touch pad, a location sensor, a gaming controller, a button, and/or detection circuitry.

In some examples, the capacitance module **200** is a mutual capacitance sensing device. In such an example, the substrate **202** has a set **204** of row electrodes and a set **206** of column electrodes that define the touch/proximity-sensitive area of the component. In some cases, the component is configured as a rectangular grid of an appropriate number of electrodes (e.g., 8-by-6, 16-by-12, 9-by-15, or the like).

As shown in FIG. 2, the capacitance module **208** includes a capacitance controller **208**. The capacitance controller **208** may include at least one of a central processing unit (CPU), a digital signal processor (DSP), an analog front end (AFE) including amplifiers, a peripheral interface controller (PIC), another type of microprocessor, and/or combinations thereof, and may be implemented as an integrated circuit, a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a combination of logic gate circuitry, other types of digital or analog electrical design components, or combinations thereof, with appropriate circuitry, hardware, firmware, and/or software to choose from available modes of operation.

In some cases, the capacitance controller **208** includes at least one multiplexing circuit to alternate which of the sets **204**, **206** of electrodes are operating as drive electrodes and sense electrodes. The driving electrodes can be driven one at a time in sequence, or randomly, or drive multiple electrodes at the same time in encoded patterns. Other configurations are possible such as a self-capacitance mode where the electrodes are driven and sensed simultaneously. Electrodes may also be arranged in non-rectangular arrays, such as radial patterns, linear strings, or the like. A shield layer (see FIG. 3) may be provided beneath the electrodes to reduce noise or other interference. The shield may extend beyond the grid of electrodes. Other configurations are also possible.

In some cases, no fixed reference point is used for measurements. The touch controller **208** may generate signals that are sent directly to the first or second sets **204**, **206** of electrodes in various patterns.

In some cases, the component does not depend upon an absolute capacitive measurement to determine the location of a finger (or stylus, pointer, or other object) on a surface of the capacitance module **200**. The capacitance module **200** may measure an imbalance in electrical charge to the electrode functioning as a sense electrode which can, in some examples, be any of the electrodes designated in either set **204**, **206** or, in other examples, with dedicated-sense electrodes. When no pointing object is on or near the capacitance module **200**, the capacitance controller **208** may be in a balanced state, and there is no signal on the sense electrode. When a finger or other pointing object creates imbalance because of capacitive coupling, a change in capacitance may occur at the intersections between the sets of electrodes **204**, **206** that make up the touch/proximity sensitive area. In some cases, the change in capacitance is measured. However, in alternative example, the absolute capacitance value may be measured.

While this example has been described with the capacitance module **200** having the flexibility of the switching the sets **204**, **206** of electrodes between sense and transmit electrodes, in other examples, each set of electrodes is dedicated to either a transmit function or a sense function.

FIG. 3 depicts an example of a substrate **202** with a first set **204** of electrodes and a second set **206** of electrodes deposited on the substrate **202** that is incorporated into a capacitance module. The first set **204** of electrodes and the

second set **206** of electrodes may be spaced apart from each other and electrically isolated from each other. In the example depicted in FIG. 3, the first set **204** of electrodes is deposited on a first side of the substrate **202**, and the second set **206** of electrodes is deposited on the second side of the substrate **202**, where the second side is opposite the first side and spaced apart by the thickness of the substrate **202**. The substrate may be made of an electrically insulating material thereby preventing the first and second sets **204**, **206** of electrodes from shorting to each other. As depicted in FIG. 2, the first set **204** of electrodes and the second set **206** of electrodes may be oriented transversely to one another. Capacitance measurements may be taken where the intersections with the electrodes from the first set **204** and the second set **206** overlap. In some examples, a voltage may be applied to the transmit electrodes and the voltage of a sense electrode that overlaps with the transmit electrode may be measured. The voltage from the sense electrode may be used to determine the capacitance at the intersection where the sense electrode overlaps with the transmit electrode.

In the example of FIG. 3 depicting a cross section of a capacitance module, the substrate **202** may be located between a capacitance reference surface **212** and a shield **214**. The capacitance reference surface **212** may be a covering that is placed over the first side of the substrate **202** and that is at least partially transparent to electric fields. As a user's finger or stylus approach the capacitance reference surface **212**, the presence of the finger or the stylus may affect the electric fields on the substrate **202**. With the presence of the finger or the stylus, the voltage measured from the sense electrode may be different than when the finger or the stylus are not present. As a result, the change in capacitance may be measured.

The shield **214** may be an electrically conductive layer that shields electric noise from the internal components of the electronic device. This shield may prevent influence on the electric fields on the substrate **202**. In some cases, the shield is solid piece of material that is electrically conductive. In other cases, the shield has a substrate and an electrically conductive material disposed on at least one substrate. In yet other examples, the shield is layer in the touch pad that performs a function and also shields the electrodes from electrically interfering noise. For example, in some examples, a pixel layer in display applications may form images that are visible through the capacitance reference surface, but also shields the electrodes from the electrical noise.

The voltage applied to the transmit electrodes may be carried through an electrical connection **216** from the touch controller **208** to the appropriate set of electrodes. The voltage applied to the sense electrode through the electric fields generated from the transmit electrode may be detected through the electrical connection **218** from the sense electrodes to the touch controller **208**.

While the example of FIG. 3 has been depicted as having both sets of electrodes deposited on a substrate, one set of electrodes deposited on a first side and a second set of electrodes deposited on a second side; in other examples, each set of electrodes may be deposited on its own dedicated substrate.

Further, while the examples above describe a touch pad with a first set of electrodes and a second set of electrodes; in some examples, the capacitance module has a single set of electrodes. In such an example, the electrodes of the sensor layer may function as both the transmit and the receive electrodes. In some cases, a voltage may be applied to an electrode for a duration of time, which changes the

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capacitance surrounding the electrode. At the conclusion of the duration of time, the application of the voltage is discontinued. Then a voltage may be measured from the same electrode to determine the capacitance. If there is no object (e.g., finger, stylus, etc.) on or in the proximity of the capacitance reference surface, then the measured voltage off of the electrode after the voltage is discontinued may be at a value that is consistent with a baseline capacitance. However, if an object is touching or in proximity to the capacitance reference surface, then the measured voltage may indicate a change in capacitance from the baseline capacitance.

In some examples, the capacitance module has a first set of electrodes and a second set of electrodes and is communication with a controller that is set up to run both mutual capacitance measurements (e.g., using both the first set and the second set of electrodes to take a capacitance measurement) or self-capacitance measurements (e.g., using just one set of electrodes to take a capacitance measurement).

FIG. 4 depicts an example of a capacitance module incorporated into a touch screen. In this example, the substrate 202, sets of electrodes 204, 206, and electrical connections 216, 218 may be similar to the arrangement described in conjunction with FIG. 3. In the example of FIG. 4, the shield 214 is located between the substrate 202 and a display layer 400. The display layer 400 may be a layer of pixels or diodes that illuminate to generate an image. The display layer may be a liquid crystal display, a light emitting diode display, an organic light emitting diode display, an electroluminescent display, a quantum dot light emitting diode display, an incandescent filaments display, a vacuum florescent display, a cathode gas display, another type of display, or combinations thereof. In this example, the shield 214, the substrate 202, and the capacitance reference surface 212 may all be at least partially optically transparent to allow the image depicted in the display layer to be visible to the user through the capacitance reference surface 212. Such a touch screen may be included in a monitor, a display assembly, a laptop, a mobile phone, a mobile device, an electronic tablet, a dashboard, a display panel, an infotainment device, another type of electronic device, or combinations thereof.

FIG. 5 depicts an example of a cross section of a capacitance module 200 where the substrate 202 may be located between a capacitance reference surface 212 and a shield 214. In this example, a first pressure sensor 500 and a second pressure sensor 502 are incorporated into the capacitance module pad 200. As depicted in this example, the pressure sensors 500, 502 may be disposed adjacent to an underside of the substrate 202. But, in other examples, the pressure sensors may be positioned at any appropriate location, including, but not limited to, adjacent the underside of the capacitance reference surface 212, adjacent the underside of the shield, another location, or combinations thereof. In examples where the pressure sensors 500, 502 are positioned under the substrate 202, pressure applied to the capacitance reference surface 212 may be transmitted through the capacitance reference surface 212 exerting a pressure on the substrate 202, which in turn applies a pressure to at least one of the pressure sensors 500, 502. In examples where the pressure sensors are positioned adjacent to the shield, the pressure applied to input surface may be transmitted to the shield, which in turn applies the pressure to the pressure sensors. This pressure may be measured by the pressure sensors 500, 502 to determine the value of the pressure. In this example, the first pressure sensor 500 is spaced apart from the second pressure sensor 502 at a distance along a

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length, width, and/or another dimension of the capacitance reference surface 212, which may allow the first pressure sensor 500 and the second pressure 502 to detect different levels of pressure depending on the location where the pressure input is made on the capacitance reference surface 212. In some cases, those pressure sensors that are closer to the location where the pressure input is made can detect a greater pressure force than the pressure sensor that is located farther away. The differing pressure values may help determine where the pressure input is made.

While this example is depicted with a pressure sensor incorporated into a capacitance module with a capacitance sensor, in other examples, the pressure sensors are not incorporated with a capacitance sensor. Further, any appropriate type of pressure sensor may be used in accordance with the principles described herein. For example, a non-exhaustive list of suitable pressure sensors includes, but is not limited to, piezoelectric sensors, magnostriuctive sensors, potentiometric pressure sensors, inductive pressure sensors, capacitive pressure sensors, strain gauge pressure sensors, variable reluctance pressure sensors, other types of pressure sensors, or combinations thereof.

In some examples, the pressure sensor may also include an ability to provide haptic feedback. For example, a piezoelectric device may be used as both a pressure sensor and as a haptic device. When the piezoelectric material is compressed due to the application of pressure through the capacitance reference surface, the piezoelectric material may produce an electric signal with can be detected by a controller. In some cases, the controller may produce an electric signal that is sent to the piezoelectric material to cause the piezoelectric material to expand, contract, and/or vibrate. The vibrations from the piezoelectric material may cause the capacitance reference surface to vibrate. This vibration may communicate a haptic signal to the user. However, in some examples, the pressure sensors are not configured to provide a haptic signal.

FIG. 6 depicts an example of a reference surface 600. In this example, a first pressure sensor 602 and a second pressure sensor 604 are located adjacent to the reference surface 600. In this example, the first pressure sensor 602 and the second pressure sensor 604 are not incorporated into a stack having a capacitance sensor.

FIGS. 7-10 depict examples of pressure sensors depicted on an underside 700 of a reference surface 702. In the example of FIG. 7, the reference surface 702 has a rectangular shape and pressure sensors 704, 706, 708, 710 are positioned in each of the corners 712, 714, 716, 718. In the example of FIG. 8, just a first pressure sensor 704 is depicted on a first side 720, and a second pressure sensor 706 is depicted on a second side 722 of the input surface 702. In the example of FIG. 9, the pressure sensors 704, 706, 708, 710 are depicted in the center of the first side 720, the second side 722, the third side 724, and the fourth side 726. In the examples of FIG. 10, the pressure sensors 704, 706, 708, 710 are depicted towards the center of the input surface and away from the edges and corners of the input surface 702.

While the examples in FIGS. 7-10 are described with reference to a specific number of pressure sensors, any appropriate number of pressure sensors may be disposed adjacent to the input surface. For example, the number of pressure sensors may include one pressure sensor or multiple pressure sensors. While the examples depicted above are described with reference to specific patterns and locations for the pressure sensors, other arrangements are contemplated including, but not limited to, symmetric distribu-

tion of sensors, an asymmetric distribution of sensors, other distributions and patterns of sensors, or combinations thereof.

FIG. 11 depicts an example of a capacitance module 1100 in accordance with the disclosure. In this example, the capacitance module 1100 includes a capacitance reference surface 1101, a first electrode layer 1102, a second electrode layer 1103, a shield layer 1104, a component layer 1105, and haptic motors 1106. While six elements of the capacitance module 1100 are identified in this example, in other examples, a capacitance module may include more or fewer elements. For example, in some examples, the capacitance module does not include a capacitance reference surface. Also, as an example, the electrodes may be combined onto one layer (either the electrodes are position on different sides of the layer or in some cases, the electrodes may be positioned on the same side of the same layer).

It should also be noted that while the elements in the capacitance module 1100 are arranged relative to other elements in the capacitance module, in other examples, the relative arrangement of elements may differ. For example, while the haptic motors 1106 are located adjacent to the component layer 1105 in this example, in another example, haptic motors may be located in between a first electrode layer and a second electrode layer, in between an electrode layer and a capacitance reference surface, in between a shield layer and a component layer, or in another location.

A user may interact with the capacitance module by touching the capacitance reference surface 1101 with a finger, stylus, or other input method. The capacitance reference surface 1101 may be made out of glass, plastic, another material, or combinations thereof.

The first electrode layer 1102 may be located adjacent to the capacitance reference surface 1101. The first electrode layer 1102 includes a first set 1107a of electrodes arranged along the width of the layer. The electrodes in the first set 1107a may be transmit electrodes, sense electrodes, another type of electrodes, or combinations thereof. The electrodes in the first set 1107a may be made of copper, gold, another conductive material, or combinations thereof.

The second electrode layer 1103 may be located adjacent to the first electrode layer 1102. The second electrode layer 1103 includes a second set 1107b of electrodes arranged along the length of the layer. The electrodes in the second set 1107b of electrodes may be transmit electrodes, sense electrodes, another type of electrodes, or combinations thereof. The electrodes in the second set 1107b may be made of copper, gold, another conductive material, or combinations thereof.

The first set 1107a of electrodes and second set 1107b of electrodes may be arranged transverse from each other on their respective layers. Together, the first electrode layer 1102 and the second electrode layer 1103 form a mutual capacitance sensor. Other arrangements of the first and second sets of electrodes are possible to form a mutual capacitance sensor.

The mutual capacitance sensor formed by the first electrode layer 1102 and second electrode layer 1103 may be used to identify user input with the capacitance module. When a user touches the capacitance reference surface 1101 with a finger, stylus, or other input method, the mutual capacitance sensor senses the input. While the capacitance module 1100 includes a mutual capacitance sensor in this example, in other examples, a capacitance module may include a self-capacitance sensor, or another type of capacitance sensor.

The shield layer 1104 may be located adjacent to the second electrode layer 1103. The shield layer 1104 is constructed to block electrical signals that may interfere with the operation of the mutual capacitance sensor formed by the first electrode layer 1102 and second electrode layer 1103. The shield layer 1104 may be made of aluminum, fiberglass, another appropriate shielding material, or combinations thereof.

The component layer 1105 may be located adjacent to the shield layer 1104. The component layer 1105 may include a printed circuit board. The component layer 1105 may contain individual components 1108 that are used to operate the touchpad. Components may include but are not limited to a central processing unit (CPU), a digital signal processor (DSP), analog front end (AFE), an amplifier, a peripheral interface controller (PIC), another type of microprocessor, an integrated circuit, a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a combination of logic gate circuitry, other types of digital or analog electrical components, or combinations thereof.

Haptic motors 1107 may be located adjacent to the component layer 1105. When activated, the haptic motors 1107 may vibrate, creating haptic feedback which may be felt through the capacitance reference surface 1101 by the user.

While the capacitance module 1100 includes four haptic motors 1107 in this example, in other examples, a capacitance module may include a different number of haptic motors. Additionally, while the haptic motors 1107 are arranged in the corners of the capacitance module 1100, in other examples, haptic motors may be located differently in a capacitance module. For example, a capacitance module may have a single haptic motor located underneath the center of the capacitance reference surface, two haptic motors located underneath the right and left side of the capacitance reference surface, two haptic motors located underneath the top and bottom sides of the capacitance reference surface, etc.

The haptic motors 1107 may be eccentric rotating mass (ERM) vibration motors, linear resonance actuator (LRA) motors, piezoelectric actuator motors, another type of haptic motor, or combinations thereof. In examples, where the haptic motors 1107 are piezoelectric actuator motors or certain other types of haptic motor, the haptic motors may also be as pressure sensors.

In some cases, the haptic motors 1107 may deliver a haptic feedback by vibrating all at once. In other cases, the haptic motors may activate at different times. Activating individual haptic motors one after another may reduce energy costs, improve a localized sensation of haptic feedback on the capacitance reference surface 1101, and minimize vibrations that are felt outside of the capacitance module 1100.

Activating individual haptic motors with signals of varying characteristics may reduce energy cost. In cases where a capacitance module creates haptic feedback by activating all haptic motors simultaneously each with a full strength, the haptic feedback may be felt throughout the entirety of a capacitance reference surface and perhaps felt by other components outside of the capacitance module. Haptic feedback may only be desired and/or perceived by a user at the location of user input on the reference capacitance surface. In some cases, it may be desirable to limit the individual haptic signals so that collective haptic signal is only limited to a sub-portion of the surface area of capacitance reference surface, such as the area equal to one or two fingertips. In some cases, any energy that generates a haptic feedback that

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can be felt by the user outside of such a limited region of the reference capacitance surface may be considered wasted energy. In contrast, if only a portion of the total haptic motors **1107** in the capacitance module **1100** are activated to produce a haptic feedback, the haptic feedback may only be perceived at or near the location of user input, saving energy and creating a more localized sensation of haptic feedback. In addition, activating only a portion of haptic motors may reduce vibrations felt outside of the capacitance module **1100**.

If only a portion of the total haptic motors **1107** are activated, the capacitance reference surface **1101** may vibrate with a weaker vibration than if all the haptic motors were activated, leading to a weaker sensation of haptic feedback. To offset this problem, haptic motors may be activated at different times. FIG. **12a** depicts an example of haptic feedback in accordance with the disclosure. In this example, a finger **1207** touches a capacitance reference **1200** at a location of user input **1205**. A first haptic motor **1201** and a second haptic motor **1202** are located underneath the capacitance reference surface **1200**. The first haptic motor **1201** generates a first haptic signal **1203**, and the second haptic motor generates a second haptic signal **1204**. The first haptic signal **1203** and the second haptic signal **1204** may have amplitudes such that the vibrations they create in the capacitance reference surface **1200** are imperceptible to a user on their own, but when these signals are added to each other at the location of user input **1205**, the two signals generate a haptic feedback **1207** which is perceptible to a user. In some cases, the first and the second haptic signals may constructively interfere with each other within the limited region of the user input thereby amplifying the signal. Such constructive interference may allow for even weaker haptic signals to be generated by the haptic motors while still having the effect of having generating larger signals.

Since the distance between the first haptic motor and the location of the user input is different than the distance between the second haptic motor and the location of the user input, one of the two haptic signals will arrive at the user input location before the other if both haptic motors send their signals at the same time. However, if the haptic motor farthest away from the user input's location is activated first the farther haptic signal may travel to target location while the closer haptic motor is still inactivated. The second haptic motor may be activated at the time necessary for the second haptic signal to arrive at the target location as the same time as the first haptic signal. This inactivity period between the activation of the first haptic motor and the second haptic motor may be an activation delay for the closer haptic motor. The activation delay may also save energy by preventing one of the signals to arrive at the user input location before the other thereby wasting energy that is not needed to create the desired haptic effect.

FIG. **12b** depicts an example of a voltage graph in accordance with the disclosure. In this graph, the voltage of the first haptic motor **1201** is represented by the line **1208**, and the voltage of the second haptic motor **1202** is represented by the line **1209**. The voltage is depicted with respect to time.

The first haptic motor **1201** and second haptic motor **1202** may be activated at different times. In this example, the first haptic motor **1201** is activated at t_1 and deactivated at t_3 . The second haptic motor **1202** is activated at t_2 and deactivated at t_4 . In other words, the first haptic motor **1201** generates the first haptic signal **1203** before the second haptic motor **1202** generates the second haptic signal **1204**. Determining

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when to activate a haptic motor may be based on the distance between the haptic motor and the desired location of haptic feedback. Because a haptic signal takes a finite time to travel through a capacitance module to the location of haptic feedback, a haptic motor that is further away from the location of haptic feedback may be activated before a haptic motor that is closer to the location of haptic feedback so that their respective haptic signals reach the location of haptic feedback at the same time. In this example, the first haptic motor **1201** is further away from the location of user input **1205** than the second haptic motor **1202**, so the first haptic motor is activated before the second.

The first haptic motor **1201** and second haptic motor **1202** may be activated with different voltages. In this example, the first haptic motor **1201** is activated with a voltage of v_2 and the second haptic motor **1202** is activated with a voltage of v_1 . The voltage of a haptic motor may correspond to the amplitude of the haptic signal generated by the motor. A higher voltage may correspond to a higher amplitude, and a lower voltage may correspond to a lower amplitude. Because a haptic signal may dissipate and decrease in amplitude as it travels further from a haptic motor, haptic motors that are further from a desired location of haptic feedback may be activated with higher voltages than haptic motors that are closer to the desired location of haptic feedback. In this example, the first haptic motor **1201** is further away from the location of user input **1205** than the second haptic motor **1202**, so the first motor is activated with a higher voltage than the second.

FIG. **13a** depicts an example of a haptic signal in accordance with the disclosure. In this example, a finger **1207** touches the capacitance reference surface **1200** at a location of user input **1303**. The first haptic motor **1201** generates a first haptic signal **1301** and the second haptic motor generates a second haptic signal **1302**. The first haptic signal **1301** and the second haptic signal **1302** generate a haptic feedback **1304** at the location of user input **1303**. In this example, the first haptic motor **1201** is closer to the location of user input **1303** than the second haptic motor **1202**.

FIG. **13b** depicts an example of a voltage graph in accordance with the disclosure. In this example, the voltage of the first haptic motor **1201** is represented by the line **1305**, and the voltage of the second haptic motors **1202** is represented by the line **1306**. The voltage is depicted with respect to time. In this example, the first haptic motor is activated at t_2 and deactivated at t_4 with an activation voltage of v_1 . The second haptic motor is activated at t_1 and deactivated at t_3 with an activation voltage of v_2 .

FIG. **14a** depicts an example of a haptic feedback in accordance with the disclosure. In this example, the finger **1207** touches the capacitance reference surface **1200** at a location of user input **1403**. The first haptic motor **1201** generates a first haptic signal **1401** and the second haptic motor **1202** generates a second haptic signal **1402**. The first haptic signal **1401** and the second haptic motors **1402** interfere at the location of user input **1403** to form a haptic feedback **1404**.

FIG. **14b** depicts an example of a voltage graph in accordance with the disclosure. In this example, the voltage of the first haptic motor **1201** is represented by the line **1405**, and the voltage of the second haptic motor **1202** is represented by the line **1406**. In this example, the voltage times and activation voltages for the first haptic motor **1201** and the second haptic motor **1202** are the same. In some examples, it may be desirable for the first haptic signal and the second haptic signal to have equal voltage strengths when the location of user input **1403** is equidistant from both

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of the motors. In the illustrated example, both the first haptic motor **1201** and the second haptic motor **1202** are activated at t_1 and deactivated at t_2 with an activation voltage of v_1 .

A haptic signal generated by one haptic motor may have a different phase, amplitude, wavelength, or frequency than a haptic signal generated by another haptic motor. FIGS. **15-18** depict examples of displacement graphs of different haptic signals that depict the differences of waveform characteristics between the different haptic motors that contribute to the overall haptic effect at the user input location.

FIG. **15** depicts an example of a displacement graph **1500** including a first haptic signal **1501** and a second haptic signal **1502**. In this example, the first haptic signal **1501** and second haptic signal **1502** share the same phase, wavelength, and frequency, but the first haptic signal has a greater amplitude than the second haptic signal.

FIG. **16** depicts an example of a displacement graph **1600** including a first haptic signal **1601** and a second haptic signal **1602**. In this example, the first haptic signal **1601** and second haptic signal **1602** share the same wavelength, frequency, and amplitude, but the two signals have different phases.

FIG. **17** depicts an example of a displacement graph **1700** including a first haptic signal **1701** and a second haptic signal **1702**. In this example, the first haptic signal **1701** and the second haptic signal **1702** have the same amplitude and phase, but they have different wavelengths and frequencies.

FIG. **18** depicts an example of a displacement graph **1800** including a first haptic signal **1801**, a second haptic signal **1802**, and a third haptic signal **1803**. In this example, each of the three haptic signals **1801, 1802, 1803** has a different phase, amplitude, wavelength, and frequency.

FIG. **19** depicts an example of pre-programmed input regions in accordance with the disclosure. A capacitance reference surface **1900**, a mutual capacitance sensor **1901**, and a haptic layer **1902** are depicted. A dotted grid over each element represents a set of input regions **1903**. The mutual capacitance sensor **1901** may include a first set **1904a** and a second set **1904b** of electrodes. The haptic layer **1902** includes a set of haptic motors **1905**.

In some examples, a capacitance module may determine which haptic motors to activate and what order to activate the haptic motors in response to determining the user input location. While this approach may provide haptic feedback that is accurate to the desired location of haptic feedback, calculating the distances of haptic motors for every appropriate input may be resource intensive and may result in the capacitance module performing more slowly. To solve this problem, input regions may be pre-programmed in a capacitance module. If the location of user input falls within a pre-programmed region, a capacitance module may select a predetermined timing and predetermined waveform characteristics to be activated by the haptic motors from stored pre-programmed instructions. Such preprogrammed instructions may be stored in the memory of the controller in the capacitance module, other memory associated with a device in which the capacitance module is incorporated, memory associated with a networked device, memory associated with a cloud device, other memory, or combinations thereof. This system may present a trade-off of haptic feedback precision in exchange for lower processing costs.

FIG. **20** depicts an example of a haptic feedback in accordance with the disclosure. In this example, a user input **2002** on the capacitance reference surface **1900** is within an input region **2001** of the set of input regions **1903**. On the haptic layer **1902**, a first haptic motor **2003** generates a first haptic signal **2005** and a second haptic motor **2004** generates

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a second haptic signal **2006**. The first haptic signal **2005** and the second haptic signal **2006** interfere at the location of user input **2002** to create a haptic feedback **2007**.

The first haptic signal **2005** and second haptic signal **2006** may have different amplitudes, phases, wavelengths, and frequencies as determined by the pre-programmed instructions associated with the input region **2001**. The timing of the activation of the first haptic motor **2003** and second haptic motor **2004** may be determined by the pre-programmed instructions associated with the input region **2001**. In this example, the first haptic signal **2005** has a greater wavelength than the second haptic signal **2006** because the first haptic motor **2003** is located further away from the input region **2001** than the second haptic motor **2004**.

While in this example the haptic feedback **2007** is created from two haptic signals **2005, 2006**, in other examples, a haptic feedback may be created from a different number of haptic signals. For example, a haptic feedback may be created from just one haptic signal, two haptic signals, three haptic signals, four haptic signals, another number of haptic signals, or combinations thereof.

FIG. **21** depicts an example of a haptic feedback in accordance with the disclosure. In this example, a user input **2102** on the capacitance reference surface **1900** is within an input region **2101** of the set of input regions **1903**. On the haptic layer **1902**, a first haptic motor **2103** generates a first haptic signal, a second haptic motor **2104** generates a second haptic signal **2108**, a third haptic motor **2105** generates a third haptic signal **2109**, and a fourth haptic motor **2106** generates a fourth haptic signal **2110**. The first haptic signal **2107**, second haptic signal **2108**, third haptic signal **2109**, and fourth haptic signal **2110** interfere at the location of user input **2102** to create a haptic feedback **2111**.

The haptic signals **2107-2110** may each have different amplitudes, phases, wavelengths, and frequencies as determined by the pre-programmed instructions associated with the input region **2101**. The timing of the haptic signals **2107-2110** may be determined by the pre-programmed instructions associated with the input region **2101**. The pre-programmed instructions which execute the activation of the haptic motors **2103-2106** may be programmed based, in part, on the distance between the input region **2101** and each of the haptic motors.

FIG. **22** depicts an example of a user input on a capacitance module in accordance with the disclosure. In this example, the capacitance module includes a capacitance reference surface **2200**, a capacitive sensor layer **2201**, and a haptic layer **2202**. While three elements are identified in the capacitance module in this example, in other examples, a capacitance module may include more or less elements.

The capacitance reference surface **2200** is located adjacent to the capacitive sensor layer **2201**. A user may interact with the capacitance module by touching the capacitance reference surface **2200** with a finger or other input method. In this example, a location of user input **2203** is identified on the capacitance reference surface **2200**.

The capacitive sensor layer **2201** is located in between the capacitance reference surface **2200** and the haptic layer **2202**. The capacitive sensor layer **2201** contains a mutual capacitance sensor **2204**. While in this example, the capacitive sensor layer **2201** includes a mutual capacitance sensor **2204**, in other examples, a capacitive sensor layer **2201** may include a different type of capacitive sensor. For example, a capacitive sensor layer may include a mutual capacitance sensor, a self-capacitance sensor, another type of capacitive sensor, or combinations thereof.

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The mutual capacitance sensor **2204** may be used to detect user input on the capacitance reference surface **2200** of the capacitance module. The capacitance module may use the information collected by the mutual capacitance sensor **2204** to determine the location and timing of a haptic feedback response.

The haptic layer **2202** is located adjacent to the capacitive sensor layer **2201**. The haptic layer **2202** includes five haptic motors. While five haptic motors are identified in the haptic layer **2202**, in other examples, a capacitance module may incorporate a different number of haptic motors.

In this example, a first haptic motor **2205** generates a first haptic signal **2208**, a second haptic motor **2206** generates a second haptic signal **2209**, and a third haptic motor **2207** generates a third haptic signal **2210**. The first haptic signal **2208**, second haptic signal **2209**, and third haptic signal **2210** interfere at the location of user input **2203** to create a haptic feedback **2211**.

In this example, the location of user input **2203** is equidistant from the first haptic motor **2205**, second haptic motor **2206**, and third haptic motor **2207**. As shown in FIGS. **23a-23e**, the first haptic signal **2208**, second haptic signal **2209**, and third haptic signal **2210** may each have different characteristics such that they are imperceptible on their own, and perceptible at the location of user input **2203** where the three signals interfere with each other to create the haptic feedback **2211**.

FIG. **23a** depicts an example of a displacement versus distance graph in accordance with the disclosure. In this example, the first haptic signal **2208** depicts an example of reference surface displacement versus linear distance from the location of user input **2203**. The vertical axis represents reference surface displacement. The horizontal axis represents linear distance from the location of user input **2203** to the first haptic motor **2205**. The location of user input **2203** is represented as $x=0$. The dashed lines on the top and bottom of the graph represent a perception threshold **2301**. If the reference surface displacement of a haptic signal exceeds the perception threshold **2301**, that haptic signal may be felt by a user. In the example FIG. **23a**, the first haptic signal **2208** does not exceed the perception threshold **2301**, therefore, it is possible that the first haptic signal may not be felt by the user.

FIG. **23b** depicts an example of a displacement versus distance graph in accordance with the disclosure. In this example, the second haptic signal **2209** is depicted as an example of reference surface displacement versus linear distance from the location of user input **2203**. In this example, the second haptic signal **2209** does not exceed the perception threshold **2301**, therefore, it is possible that in some cases the second haptic signal may not be felt by the user.

FIG. **23c** depicts an example of a displacement versus distance graph in accordance with the disclosure. In this example, the third haptic signal **2210** is depicted as an example of reference surface displacement versus linear distance from the location of user input **2203**. In this example, the third haptic signal **2210** does not exceed the perception threshold **2301**, therefore, it is possible that the third haptic signal may not be felt by the user.

It should be noted that while the first, second, and third haptic signals **2208-2210** are equidistant from the location of user input **2210**, they do not share the same characteristics. For example, the amplitude of the third haptic signal **2210** has a greater amplitude than either the first haptic signal **2208** or second haptic signal **2209**. The second haptic signal **2209** has a greater frequency and shorter wavelength than

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either the first haptic signal **2208** or third haptic signal **2210**. Varying the characteristics of different haptic signals may be suited to prevent a haptic feedback from being perceived in a location other than the desired feedback location. By varying the characteristics of haptic signals, the perception of feedback may be reduced at undesired locations and increased at or near desired locations.

FIG. **23d** depicts an example of a displacement versus distance graph in accordance with the disclosure. In this example, the first haptic signal **2208**, second haptic signal **2209**, and third haptic signal **2210** depicted on the same graph superimposed on each other.

FIG. **23e** depicts an example of a displacement versus distance graph in accordance with the disclosure. In this example, a composite signal **2302** is the sum of the first haptic signal **2208**, second haptic signal **2209**, and third haptic signal **2210**. The composite signal **2302** represents how each of the individual haptic signals **2208-2210** may constructively and destructively interfere to create the haptic feedback **2211**. As shown in FIG. **23e**, the composite signal **2302** exceeds the perception threshold **2301** at the location of user input **2302**. Therefore, the composite signal **2302** may be felt by a user at the location of user input **2302** as haptic feedback **2211**.

FIG. **24** depicts a method **2400** for delivering a haptic feedback with a capacitance module in accordance with the disclosure. This method **2400** may be performed based on the description of the devices, modules, and principles described in relation to FIGS. **1-23**. In this example, the method **2400** may include delivering a haptic feedback with a capacitance module includes obtaining **2401** a location of user input using a capacitive sensor in the capacitance module, determining **2402** when to activate a first haptic motor in the capacitance module, based on the location of the user input, determining **2403** when to activate a second haptic motor in the capacitance module based on the location of user input, generating **2406** the first haptic signal with the first haptic motor at the first determined time, and generating **2407** the second haptic signal with the second haptic motor at the second determined time. The method **2400** may optionally include determining **2404** the amplitude, wavelength, frequency, and phase of the first haptic signal. The method **2400** may optionally include determining **2405** the amplitude, wavelength, frequency, and phase of the second haptic signal based on the amplitude, wavelength, frequency, and phase of the first haptic signal.

It should be noted that while methods, systems, and devices discussed may describe implementations of capacitance modules as touchpads in laptop devices, the disclosed methods, systems, and devices may be implemented into other devices that include a capacitance module. Devices that may implement the methods, systems, and devices discussed above may include but are not limited to laptop devices, smartphone devices, video-game controller devices, kiosk devices, touchscreen devices, other devices, or combinations thereof.

For example, in cases where a video-game controller includes a capacitance module with haptic motors, the motors may deliver haptic feedback as described above. In another example, in cases where a smartphone includes a touch screen with haptic motors, the motors may deliver haptic feedback as described above.

It should be noted that the methods, systems, and devices discussed above are intended merely to be examples. It must be stressed that various embodiments may omit, substitute, or add various procedures or components as appropriate. For instance, it should be appreciated that, in alternative embodi-

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ments, the methods may be performed in an order different from that described, and that various steps may be added, omitted or combined. Also, features described with respect to certain embodiments may be combined in various other embodiments. Different aspects and elements of the embodiments may be combined in a similar manner. Also, it should be emphasized that technology evolves and, thus, many of the elements are exemplary in nature and should not be interpreted to limit the scope of the invention.

Specific details are given in the description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the embodiments.

Also, it is noted that the embodiments may be described as a process which is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure.

Having described several embodiments, it will be recognized by those of skill in the art that various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the invention. For example, the above elements may merely be a component of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered. Accordingly, the above description should not be taken as limiting the scope of the invention.

The invention claimed is:

1. A capacitance module, comprising:
a capacitive sensor;
a first haptic motor;
a second haptic motor;
wherein the first haptic motor and second haptic motor are in mechanical communication with the capacitive sensor;
controlling resources connected to the capacitive sensor and the plurality of haptic motors with memory;
wherein the memory contains programmed instructions that cause the controlling resources, when activated, to:
determine when to activate the first haptic motor in response to a user input;
determine when to activate the second haptic motor in response to the user input;
generate a first haptic signal with the first haptic motor at a first determined time; and
generate a second haptic signal with the second haptic motor at a second determined time;
wherein the first haptic signal and second haptic signal constructively interfere at the location of user input.
2. The module of claim 1, wherein the programmed instructions cause the controlling resources, when activated, to:
determine the waveform characteristics of the first haptic signal; and
determine the waveform characteristics of the second haptic signal based on the waveform characteristics of the first haptic signal.

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3. The module of claim 1, wherein the first haptic signal and second haptic signal generate a haptic feedback that is just tangible at and/or near the location of user input.

4. The module of claim 1, wherein the first haptic signal and second haptic signal are timed to arrive at the location of user input at the same time.

5. The module of claim 1, wherein:

determining when to activate the first haptic motor includes determining a distance between the user input and the first haptic motor; and

determining when to activate the second haptic motor includes determining a distance between the user input and the second haptic motor.

6. The module of claim 1, wherein the capacitive sensor includes at least one pre-defined region, wherein the first determined time and second determined time are pre-programmed based, at least in part, on the at least one pre-defined region.

7. The module of claim 1, wherein the first determined time and the second determined time have the same value.

8. The module of claim 1, wherein the capacitance module is a touchpad.

9. The module of claim 1, wherein the first haptic signal and the second haptic signal are tuned to minimize vibrations perceptible outside of the area defined by the capacitance module.

10. A method for delivering a haptic feedback with a capacitance module, comprising:

obtaining a location of user input using a capacitive sensor in the capacitance module;

determining when to activate a first haptic motor in the capacitance module based on the location of the user input;

determining when to activate a second haptic motor in the capacitance module based on the location of the user input;

generating a first haptic signal with the first haptic motor at the first determined time; and

generating a second haptic signal with the second haptic motor at the second determined time;

wherein the haptic feedback comprises the first haptic signal and second haptic signal;

wherein the first haptic signal and second haptic signal constructively interfere at the location of user input.

11. The method of claim 10, further comprising:

determining the waveform characteristics of the first haptic signal; and

determining the waveform characteristics of the second haptic signal based on the waveform characteristics of the first haptic signal.

12. The method of claim 10, wherein the first haptic signal and second haptic signal generate a haptic feedback that is just perceptible at and/or near the location of user input.

13. The method of claim 10, wherein the first haptic signal and second haptic signal are timed to arrive at the location of user input at the same time.

14. The method of claim 10, wherein:

determining when to activate the first haptic motor includes determining a distance between the user input and the first haptic motor; and

determining when to activate the second haptic motor includes determining a distance between the user input and the second haptic motor.

15. The method of claim 10, wherein:

the capacitive sensor includes at least one pre-defined region; and

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the first determined time and second determined time are pre-programmed based, in part, on the pre-defined region.

16. The method of claim **10**, wherein the capacitance module is a touchpad.

17. A computer-program product for generating a haptic feedback using a capacitance module, the computer-program product comprising a non-transitory, computer readable medium storing instructions executable by a processor to:

obtain a location of user input using a capacitive sensor in the capacitance module;

determine when to activate a first haptic motor in the capacitance module based on the location of the user input;

determine when to activate a second haptic motor in the capacitance module based on the location of the user input;

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generate a first haptic signal with the first haptic motor at the first determined time; and

generate a second haptic signal with the second haptic motor at the second determined time;

5 wherein the haptic feedback comprises the first haptic signal and second haptic signal;

wherein the first haptic signal and second haptic signal constructively interfere at the location of user input.

18. The computer-program product of claim **17**, the medium storing further instructions executable by a processor to:

determine the waveform characteristics of the first haptic signal; and

15 determine the waveform characteristics of the second haptic signal based on the waveform characteristics of the second haptic signal.

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